



INVESTOR REPORT AND COMPARISON PACKET

A companion that knows when to talk

Voice-first, location-aware storytelling for people who walk and drive. What was built, what was measured, what is still unproven.

Status of the evidence. The product has had **zero real users**, zero runs on a real device and zero runs against a live provider. Performance figures come from real code against fake providers with simulated latency; dollar figures are list-price estimates; market figures are bottom-up assumptions. Every number in this report carries a label saying which.

Telvey is a working brand name, not legally cleared; its domains are unregistered.

Report date 2026-10-04 · Repository branch `build/mvp` · Commit `51a787e` plus uncommitted report files

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1. Executive summary

PART I · THE OPPORTUNITY AND THE EVIDENCE IN ONE PAGE

What it is. Telvey (a *working* brand, not legally cleared) is a voice-first companion for people who are walking or driving. It notices what is around or ahead, tells a short story about it that is checked against approved source facts (Wikipedia and Wikidata; no source is shown to the listener today), and stays silent when nothing is worth saying. A user can interrupt with a question (“Where can I get coffee nearby?”), see the answer on a map and rejoin the story where it stopped. Two original AI Guides, Ida and Emil, tell the same facts in different voices.

Who it is for and where. Travellers on foot, road-trippers, long-haul truck drivers (audio-first, hands-free) and curious locals. The product is global and location-first by owner decision. New York, Chicago, San Francisco and U.S. highways are *test geographies*, not launch markets.

Why now, and why it is not obvious. Language models and speech synthesis make narration of any place affordable. The platforms already answer place questions for free (Google Maps with Gemini, Gemini in Android Auto, Siri AI in iOS 27, ChatGPT voice). As of the research date, 2026-09-27, none was found narrating proactively. That gap is the product, and it is also the main risk, because a platform could close it with one setting.

What was built (all in this repository, no users yet). A TypeScript monorepo of about 28,000 lines of source plus about 5,000 lines of tests: a decision core, provider adapters with deterministic fakes, a Fastify API on Postgres and Redis, a Next.js website, WebApp and admin console, and an Expo mobile app that bundles but has never run on a device. **460 automated tests pass.** A 27-scenario acceptance benchmark reports **13 PASS, 8 PARTIAL, 0 FAIL, 6 NOT RUN** (PASS means demonstrated on synthetic fixtures with fake providers; two self-set cost budgets are missed, see section 15.3).

Who built it, and who checked it. The code, tests, fixtures, replay traces, benchmarks, cost model and this report were produced by AI coding agents for a single product owner. No independent human code, test or security review has been done. Tests and fixtures were written by the same agents as the code, so passing tests show internal consistency, not correctness in the world.

Evidence level. The product has had zero real users, zero real-device runs and zero live-provider runs. Every performance number below is a **MEASURED-harness** or **MEASURED-replay** figure from real code running against fake providers with simulated latency; every dollar figure is an **ESTIMATED** list-price model; every market figure is a bottom-up **ASSUMED** model.

Headline number	Value	Label
Tests passing (core, providers, replay, client, API, mobile)	460 of 460	MEASURED-tests
Acceptance scenarios	13 PASS, 8 PARTIAL, 0 FAIL, 6 NOT RUN	MEASURED-tests / -replay / -harness
Approved moment to first audio, p50 / p95 (target 2.5 s / 4.5 s)	1,663 ms / 2,683 ms	MEASURED-harness , simulated providers
30-minute walk, default mix, no shared cache	\$0.306	ESTIMATED
30-minute highway drive, default / with tiered economy voice	\$0.0167 / \$0.0040	ESTIMATED
Long-haul driver month (176 h, all sparse highway; \$7.12 at 10% urban hours), tiered voice (economy on the highway), 0% / 50% shared cache	\$3.61 / \$2.96	ESTIMATED ; 50% is ASSUMED
Silence on the I-40 replay (105 min)	98.7% (about 96% with model-length stories), longest gap 32 min	MEASURED-replay , synthetic trace; 96% is DERIVED
Consumer TAM / SAM / year-3 SOM (three sized segments)	\$5.61B / \$1.86B / \$6.5M	ASSUMED , bottom-up

Commercial thesis. Guest-first freemium, a \$9.99 trip pass and a \$39.99 annual plan (**HYPOTHESIS**), anchored on Autio and Roadtrippers), then a driver plan once highway cost is proven. A paying Explorer user has a modelled 41% variable margin. The free tier as originally specified is not affordable.

Largest unresolved risks, in order.

- 1. **Platform risk.** Google, Apple or OpenAI can add proactive narration to an assistant that is already installed and free. Our own estimate puts Google’s chance of closing the gap within 12 months at medium-high.
- 2. **Cost and cache.** Text-to-speech is 94% to 99% of variable cost in a plain session. The driver plan only works with an economy voice and a shared story cache; the production cache hit rate is **NOT YET MEASURED** , and the economy voices are unauditioned.
- 3. **Nothing has met reality.** No live provider, device, road, network or human listener has touched the product. Wikimedia is unreachable from this sandbox, so live discovery is untested; realtime voice, iOS builds and any public deployment are blocked on credentials.
- 4. **Platform terms and brand.** Google Places terms limit caching and require a Google map; that display rule is documented but not yet enforced in code. The name Telvey is not legally cleared and the domains are unregistered.

Decisive next experiment. Supply the blocked credentials, run the realtime voice benchmark and a real-road field test, then put the product in 50 to 200 hands to measure cache hit rate, cost per session and whether anyone returns. Section 19 shows what each funding step is meant to prove.

2. Customer problem and target market

Honesty note. No interviews, surveys or pilots have been run. The segments and jobs below are hypotheses assembled from public data and from competitor traction. They are the first things a beta must test.

2.1 The problem

Most of what is interesting about a place is invisible from the sidewalk or the driver’s seat. People who want to know more have three poor options today. They can stop and search, which breaks the moment. They can buy a scripted audio tour, which only works on its authored route. Or they can ask an assistant, which answers only if they already know what to ask. Nothing decides *for* them what is worth saying about the thing in front of them, and nothing knows when to stay quiet.

2.2 Segments and jobs to be done

Segment	Job to be done	Context where value is highest	Why it matters commercially	Evidence quality
Traveller on foot	“Tell me what I am looking at, briefly, and let me ask follow-ups.”	Unfamiliar dense city, first visit	Pays per trip: Autio 14-day \$14.99, Narrativ €14.99 per 3 days	Competitor pricing (sourced)
Road-tripper	“Make the miles interesting without me touching the phone.”	Interstates and scenic routes, hours at a time	Repeat buyers; Roadtrippers Premium (\$59.99 a year) bundles Autio	Competitor pricing (sourced)
Long-haul truck driver	“Keep me company and awake, legally and safely.”	Sparse highway, up to 11 driving hours a shift, silence is often correct	2.22M heavy-truck jobs (BLS); willingness to pay unknown	Population sourced; WTP ASSUMED
Local explorer	“Show me my own city again.”	Weekend walks, commutes	High engagement, low willingness to pay	Not sized
Channel buyers (tourism boards, rental fleets, carriers, car makers)	“Offer a guide to our visitors, renters, drivers or owners.”	White-label, in-car	Small today; long cycles; Autio inside Ford and Lincoln shows OEMs buy	Competitor evidence only

2.3 Current alternatives and why they fall short

- **Scripted audio-tour apps** (Autio, VoiceMap, GuideAlong, Shaka Guide, SmartGuide, izi.TRAVEL, GPSmyCity). Strong willingness to pay (\$15 to \$40 per tour, \$36 to \$50 a year) and high ratings, but coverage ends where authoring ends.
- **Platform assistants** (Gemini in Maps and Android Auto, Siri AI, ChatGPT voice). Free and installed, good at answering, but the user has to ask. A hands-on review in February 2026 found Gemini in Maps strong on neighbourhood trivia and weak at route edits.
- **AI-native lookalikes** (RoadWhisper, Roadguide, Nextour, Narrativ, Road Trip Voice Tour Guide AI). Dynamic narration of any place, but negligible public traction (Google Play counts of 100+ to 1K+ where public; several newer entrants publish no counts)

and, as far as public descriptions show, interval or radius triggers with no stated grounding check or silence policy.

- **Radio, podcasts and silence.** In-car listening in the past month: AM/FM 73%, online audio 48%, podcasts 37% (Infinite Dial 2026, retrieved 2026-09-27). The product competes for that time.

2.4 Where the product should win

The strongest cases are those where *deciding when to speak* matters: a driver on a sparse highway (where silence is the correct behaviour: about 96% of the time with model-length stories in our replay (98.7% with the shorter template speech), on a sparse fixture where few candidates exist), a pedestrian in a dense district where dozens of places compete for attention, and a returning user who should not hear the same story twice. These are design claims backed by replay tests on synthetic traces. Whether people value them is unmeasured.

3. Market potential and competitive landscape

3.1 Competitors and traction

Competitor	Type	How it finds things	Price (sourced)	Public traction (sourced, point in time)
Google Maps with Gemini	Platform	User asks	Free	Ask Maps launched US and India 2026-03-12 with CarPlay and Android Auto; Gemini in walking and cycling navigation worldwide Feb 2026
Gemini in Android Auto	Platform	User asks	Free	Global rollout, 45 languages
Apple Siri AI (iOS 27), Maps	Platform	User asks	Free	iOS 27 released Sept 2026; no Maps narration feature found
ChatGPT voice	Platform	User asks	Freemium	Location sharing Mar 2026; CarPlay voice, audio-only
Autio	Direct, scripted	GPS-triggered library of 20,000+ stories	\$14.99 (14 days) to \$49.99 (1 year)	iOS 4.8 stars / 4.3K ratings; Play 3.5 stars, 10K+; 230K registered users (2023); in Ford and Lincoln vehicles
GuideAlong	Direct, scripted	100+ driving tours	\$14.99 to \$39.99 per tour	iOS 4.9 stars / 16K ratings
VoiceMap	Direct, scripted	2,100+ tours	\$3.99 to \$19.99 per tour	iOS 4.8 stars / 3.4K; Play 100K+
Shaka Guide, SmartGuide, izi.TRAVEL	Direct, scripted	Authored tours	Per tour / B2B €98 to €780 a month / membership, price not published	Play 100K+ / 1M+ / 1M+ (izi.TRAVEL 2.3 stars)
Le Walk	Direct, scripted	Cinematic walks	n/a	\$4.1M seed; 110K beta downloads (2025-09-19)
AI-native narrators (RoadWhisper, Roadguide, Nextour, Narrativ and others)	Direct, dynamic	Radius or interval triggers	Free to \$2.99 a month; credits	Play counts of 100+ to 1K+ where public
Waytale	Direct: GPS-triggered playback, AI narrators (live generation vs pre-authored not verified)	Passes 3 days \$3.99, 7 days \$5.99, 14 days \$9.99	First released 2025-09-03; no public ratings yet	App Store listing, checked 2026-10-04: “Companion Mode” plays stories automatically as you explore, three AI narrators; no Q&A or CarPlay listed

Table 1. Selected competitors. Full list of 19 entries (one groups five AI-native apps) in `benchmark/research/competitors.json`, retrieved 2026-09-27.

Competitor feature matrix

Sourced competitor cells (as of 2026-09-27) next to what Telvey actually has today

	Any place (dynamic)	Proactive narration	Voice Q&A	Driving mode	Looks ahead	Walking mode	Journey memory	Silence policy	Choice of voices	CarPlay / Andr. Auto	Offline	Global coverage
Telvey as built (fixtures, fakes)	Partial	Yes	Partial	Partial	Yes	Yes	Yes	Yes	Partial	No	Partial	Partial
Telvey target (not built)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Partial	Partial	Yes
Google Maps + Gemini	Yes	No	Yes	Yes	Partial	Yes	?	No	No	Yes	Partial	Yes
Gemini in Android Auto	Yes	No	Yes	Yes	Partial	No	?	No	No	Yes	No	Yes
Apple Siri AI + Maps	Partial	No	Yes	Yes	No	Partial	?	No	No	Yes	Partial	Yes
ChatGPT voice	Partial	No	Yes	Partial	No	Partial	Partial	No	No	Yes	No	Yes
Autio	No	Yes	No	Yes	Partial	No	Partial	?	No	Yes	?	Partial
VoiceMap	No	Yes	No	Partial	No	Yes	No	No	No	?	Yes	Partial
GuideAlong	No	Yes	No	Yes	No	Partial	No	No	No	Yes	Yes	Partial
Shaka Guide	No	Yes	No	Yes	No	Yes	No	No	No	?	Yes	Partial
SmartGuide	No	Yes	No	No	No	Yes	No	No	No	?	Yes	Partial
izi.TRAVEL	No	Yes	No	No	No	Yes	No	No	No	?	Yes	Partial
RoadWhisper	Yes	Yes	No	Yes	No	No	?	Partial	Partial	No	No	Yes
Nextour	Yes	Partial	Yes	No	No	Yes	?	No	Yes	?	No	Yes
Narrativ	Yes	Yes	Partial	No	No	Yes	?	No	Partial	?	Yes	Yes
Waytale (listing read 2026-10-04)	?	Yes	No	?	?	?	?	?	Yes	No	?	?

Yes (sourced for competitors)
Partial
No / not found
Unknown

Telvey as built: Yes = works end to end in tests, replay or the harness with fake providers; Partial = built but not proven live, on a device or with real providers; No = not built. Our own assessment.

Telvey target row is the aspiration from the earlier feature matrix and is NOT what exists today (CarPlay / Android Auto: not built, needs an Apple entitlement; offline: prefetch only).

Waytale (AI audio city guide, first released 2025-09-03): proactive, voices and the No cells come from its App Store listing read on 2026-10-04; whether stories are generated live is unknown. No / not found means not found in public sources, not proof of absence. Voice choice for Telvey is Partial because no voice has been heard.

Source: benchmark/research/competitors.json (featureMatrix, retrieved 2026-09-27); Telvey as-built row from results.json and docs/STATUS.md.

Figure 1. Competitor feature matrix. Competitor cells are sourced (retrieved 2026-09-27). The Telvey rows are our own assessment: “as built” counts what works in tests, replay or the harness; “target” is an aspiration and is not what exists.

Data: benchmark/research/competitors.json; benchmark/acceptance/results.json.

Three observations follow from **Figure 1**. First, proactive narration exists in many scripted apps and, per Waytale’s listing, in at least one AI-narrated app that has been live for about thirteen months, so “proactive” and “AI voices” are not differentiators; the differences are *any place*, *trajectory-ahead* reasoning, an explicit *silence policy* and *journey memory*, and each of those is a design claim not yet validated by users. Second, the as-built row is mostly “partial” because nothing has been proven live, on a device or with real providers. Third, the matrix understates how crowded the AI-native end is: search results on 2026-10-04 show at least Waytale (listing read), Guidel (tap-to-listen and image-scan guide, no ratings yet), Tourly Audio, CityTour AI, Guideline, Narriva and Votura alongside the lookalikes already tracked (the others by title only). Absence of a feature in the matrix means “not found in public sources”, not “absent”.

3.2 The big-platform risk

Actor	Shipped (sourced, 2026-09-27)	Gap left	Chance it closes the gap within 12 months (our judgment, ASSUMED)
Google	Gemini in Maps for driving, walking and cycling; Ask Maps; Gemini Live in Android Auto; landmark-based guidance	Proactive, paced story narration; silence policy; personas; cross-session memory	Medium-high
Apple	Siri AI (Gemini-based) in iOS 27; Visual Intelligence; controls CarPlay entitlements	Location-proactive narration	Low-medium

Actor	Shipped (sourced, 2026-09-27)	Gap left	Chance it closes the gap within 12 months (our judgment, ASSUMED)
OpenAI	Location sharing; ChatGPT voice in CarPlay	Background location; proactive triggers; map grounding	Medium
Car makers	Mercedes MBUX with Google; Ford and Lincoln with Auto	Cross-brand, phone-based product	Channels as much as threats

The honest reading: Q&A about a place is table stakes and free, so it cannot be the product. The research did not find a proactive narration feature in Google's 2025 and 2026 announcements, but absence of evidence in a web search is weak evidence. If a platform adds one toggle, the core proposition is commoditised and only verticals (long-haul drivers, destination partners, fleets) and execution quality remain.

3.3 Market size: a bottom-up bridge

Third-party headlines are context, not inputs. One paid report puts self-guided audio tours at \$2.86B in 2025 growing to \$5.71B by 2034 (Verified Market Reports; methodology not visible, definitions unclear). A venture survey puts consumer AI spend at \$40B in 2026 (Menlo Ventures; its own headline notes that user growth barely moved). Neither is used below.

The model multiplies a population by a usage share by a price, and labels every input as sourced or assumed.

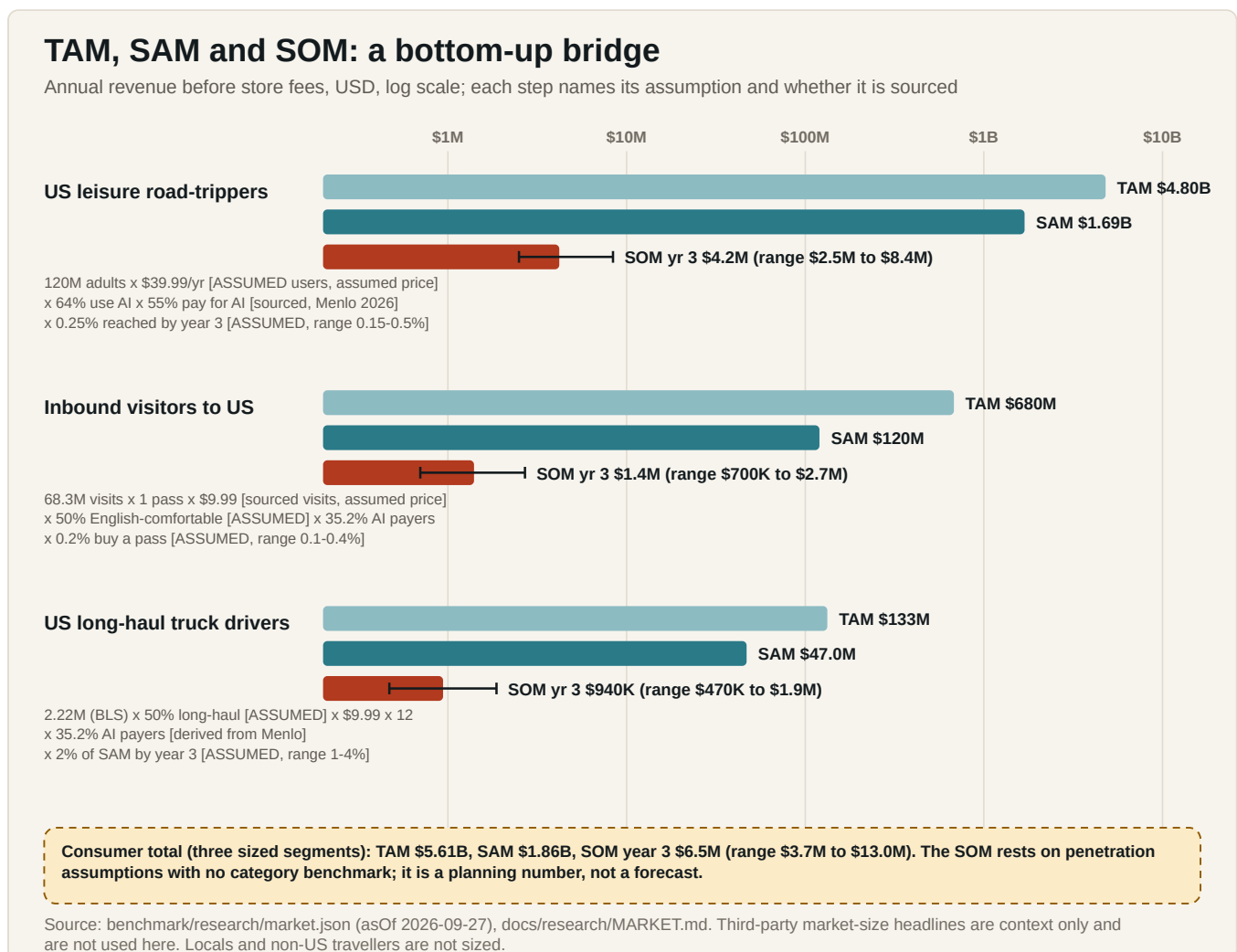


Figure 2. TAM, SAM and SOM bridge for three consumer segments, log scale, with each step's assumption.

Data: benchmark/research/market.json (asOf 2026-09-27), docs/research/MARKET.md.

Segment	TAM	SAM	SOM, year 3 (range)	Weakest assumption
US leisure road-trippers	\$4.80B (120M adults x \$39.99)	\$1.69B (x 64% use AI x 55% pay)	\$4.2M (\$2.5M to \$8.4M) at 0.25%	120M unique adults: no sourced count; must exceed 39.1M on one holiday weekend
Inbound visitors to the US	\$680M (68.3M visits x \$9.99)	\$120M	\$1.4M (\$0.7M to \$2.7M) at 0.2%	50% English-comfortable; 0.2% buy
US long-haul truck drivers	\$133M (2.22M x 50% x \$9.99 x 12)	\$47M	\$0.94M (\$0.47M to \$1.87M) at 2%	50% long-haul share is a placeholder
Consumer total	\$5.61B	\$1.86B	\$6.5M (\$3.7M to \$13.0M)	Penetration has no category benchmark

Table 1. Figures are annual revenue before store fees (net is about 85% only if Apple’s Small Business Program applies, which holds up to \$1M of proceeds; above that, standard store rates apply). Locals and non-US travellers are not sized because no defensible usage or willingness-to-pay input exists.

Read the year-3 figure as an upper bound, not a forecast. The SOM is 0.25% of the serviceable market, about 106,000 paying road-trip subscribers, roughly 46% of the registered (not paying) base Autio reports after about five years; it models no ramp, acquisition cost or churn, and the research cites about 72% first-year annual cancellation for subscription apps. The inbound-visitor segment is 0.2% of all visits (about 1.1% of its SAM) while the others are a share of SAM. The SAM filter is a US-adult proxy (AI-payer share) applied to non-US visitors and drivers, the AAA traveller count includes children, and the 68.3 million inbound visits include Canada and Mexico and count visits, not persons. A downside case at 0.05% of SAM would be about one fifth of the SOM shown.

Business-to-business segments are small on a bottom-up basis: tourism boards \$7.9M (658 member organisations x \$12,000), car rental \$13.9M, fleet driver wellbeing \$80M, vehicle makers \$48.6M, about \$150M together and mostly a distribution channel. For scale: Autio reached about 230,000 *registered* users (not payers) roughly five years after founding. The SOM above should be read as “what 0.25% of a plausible audience would pay”, not a forecast.

4. Differentiation and defensibility

Claimed advantage	Exists today?	What the evidence is	What is missing
Decides when to speak, including long silence	Built (policy; silence on a sparse fixture reflects few candidates as much as judgement, and no naive-trigger baseline has been run)	Replay on synthetic traces: 98.7% silent on I-40 (105 min) and 84% to 91% on city walks (template-length speech; see section 10.3), zero stories behind the user or passed mid-story (MEASURED-replay)	Walking silence is overstated, see section 10.3; no user has judged the timing
Looks ahead along the trajectory	Built	A5 replay: 5 stories, 0 behind, look-ahead scales with speed and significance	Real-road field run; live discovery
Facts are grounded	Built, tested with fake prose	GroundingCheck rejects unsupported numbers, years and names; 0 grounding failures across all replays	No real LLM has been checked (B-series PARTIAL)
Exact interruption and resume	Built	Coffee flow passes 36 of 36 harness sessions (one scripted scenario repeated 36 times with seeded delays, not 36 independent cases); resume at the start of the interrupted segment with a bridge phrase	Device audio behaviour not tested
Journey memory, no retelling	Built	No re-told targets in any replay; 30-day cross-session retell window (D-023)	Real usage patterns unknown
Two Guides with one set of facts	Built at brief level	Same lead fact in 31 of 31 evidence packs; verbosity 1.15 against 0.80	Live LLM prose and human rating not run (B2)
Provider independence	Built	Interfaces, ordered fallback chains, fakes; every test swaps providers	No real provider has been exercised or swapped
Cost architecture (shared story cache, tiered voice)	Built	100% body-cache hits in a one-route harness repeat	Production hit rate not measured

Which moats exist today. Honestly, none beyond execution detail. The code is copyable, the lookalike apps show how easily the concept is copied, and switching costs for users are low (journey memory is small). The defensibility case is therefore a set of *future hypotheses*:

- **Feedback loop.** Skips, “not that one” and ratings tell the system which places and angles are worth interrupting for. It needs users to exist.
- **Shared-content cost advantage.** A shared story body and shared audio let marginal cost fall as more people hear the same place. It is only a moat if real hit rates are high.
- **Distribution.** CarPlay, Android Auto, vehicle makers and tourism partners are gated by entitlements and relationships, not code.
- **Editorial quality.** Timing, grounding and silence are craft. Competitors can copy them, but only by doing the same unglamorous work.

5. Product and user experience

PART II · THE PRODUCT

5.1 What a session looks like

A visitor opens the WebApp or app. There is no account and no card: a guest token is issued and a Start card offers “Use my location” or one of four simulated trips. They pick a Guide (Ida by default), press start, and put the phone away.

From then on the product runs a loop. It reads the live context (position, heading, speed, whether the person is walking, driving or standing still, how dense the surroundings are) and asks what is worth saying about something nearby or ahead. Most of the time the answer is *nothing*, and the app says so quietly (“Quiet stretch”). When a target clears the bar, the Guide tells a short story in segments, with a spoken cue about where the thing is (“straight ahead, about 800 feet”). The listener can interrupt at any point by pressing the mic: the story stops, the question is answered from verified data, the map updates, and the story resumes where it was cut off. When the system detects driving, the screen collapses to a single status line and one large voice button.

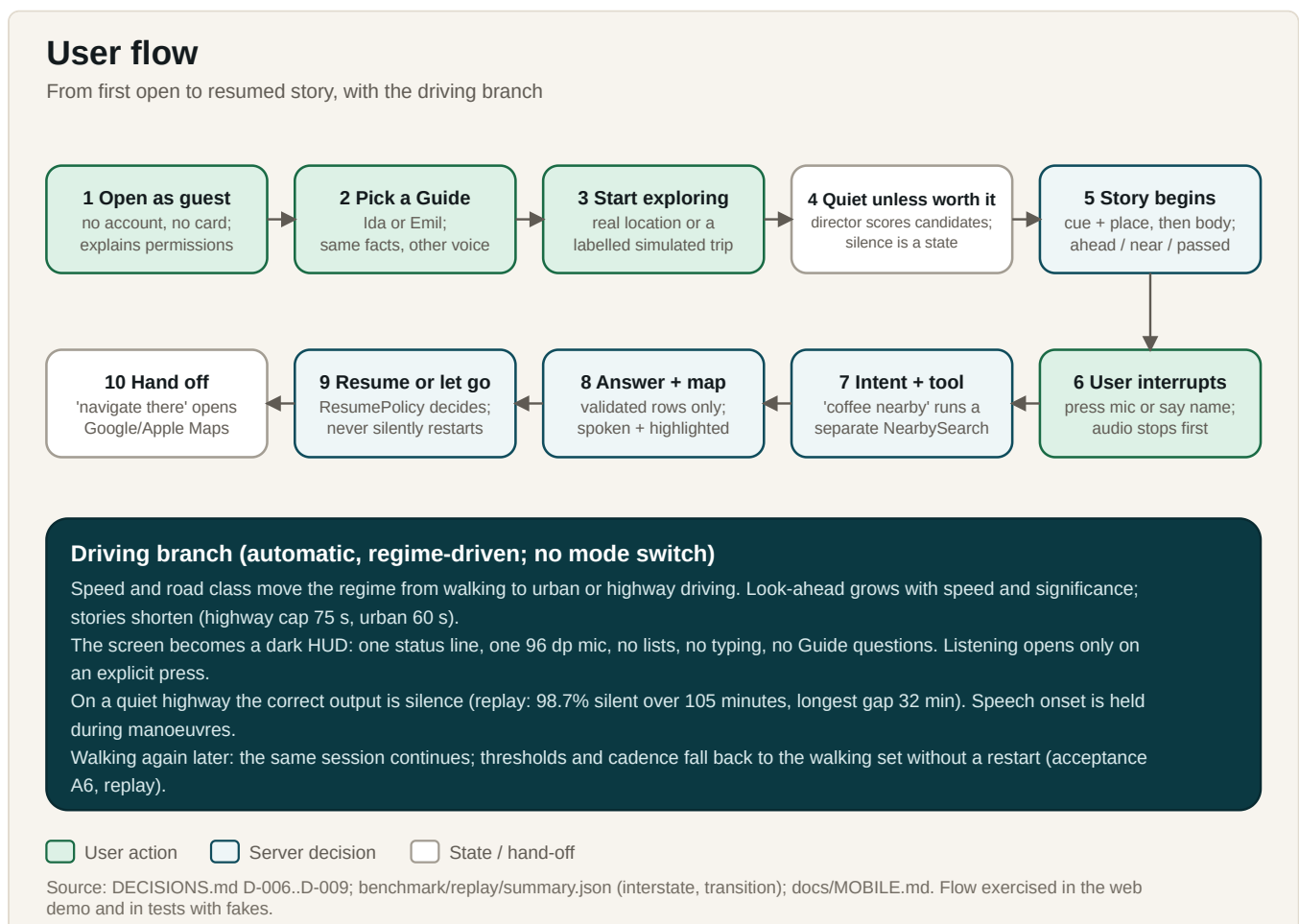


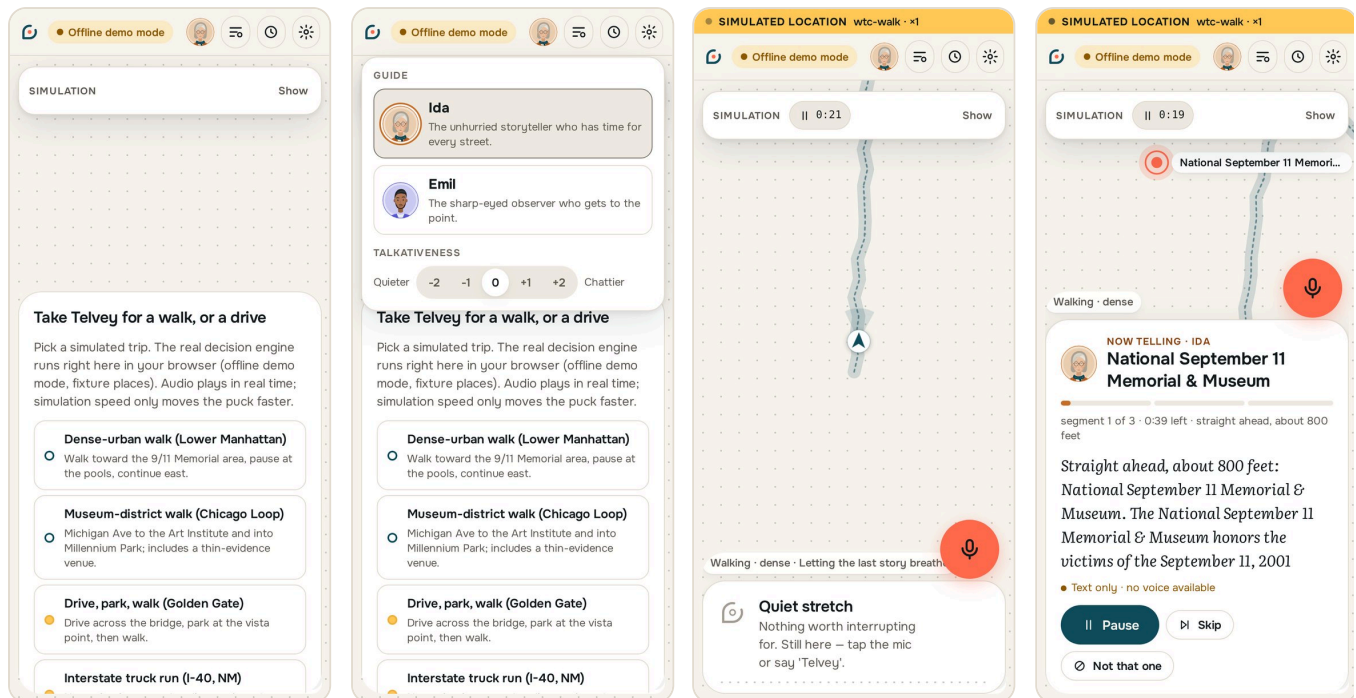
Figure 3. The guest user flow, from first open to resumed story.

Data: product design; every state shown was captured in the screenshots below.

5.2 Screen walkthrough

The twelve phone-width captures below were taken from the production build of the WebApp, running locally in offline demo mode, on 2026-10-04. Each caption states what was actually on screen; none is a mockup.

How to read these screenshots. Every capture is *offline demo mode*: the real decision code runs in the browser over fixture places and evidence. The location is simulated (an amber banner says so), the story text is template prose with no language model, there is no speech voice, and the map is a schematic overlay because tile servers are unreachable from the capture sandbox. “Cafe #NNN (synthetic)” markers are generated, not real businesses. Nothing here shows voice quality, real maps or on-device behaviour. The status label for this whole section is **MEASURED-e2e** for behaviour and **NOT RUN** for everything on a device.

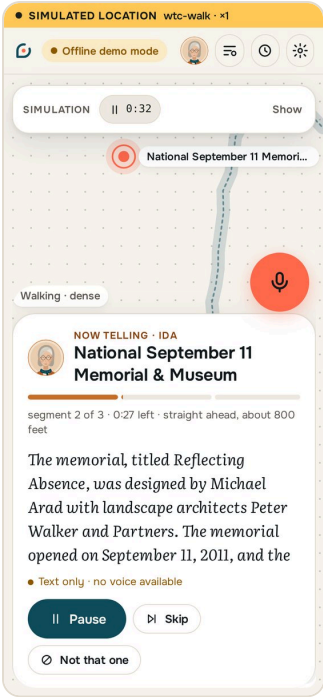


1 Guest entry. First run: no account, no card. Start card with four simulated trips and “Use my location”; chip reads “Offline demo mode”. Captured before any trip starts.

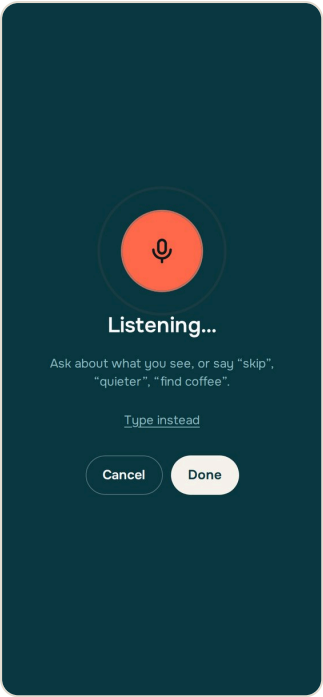
2 Guide selection. Ida selected, Emil available, talkativeness control at its default. Same facts, different voice and angle.

3 Explore, idle. After a skip, the director enforces a cadence gap: “Quiet stretch”, regime “Walking, dense”. Silence is a designed state, not an error.

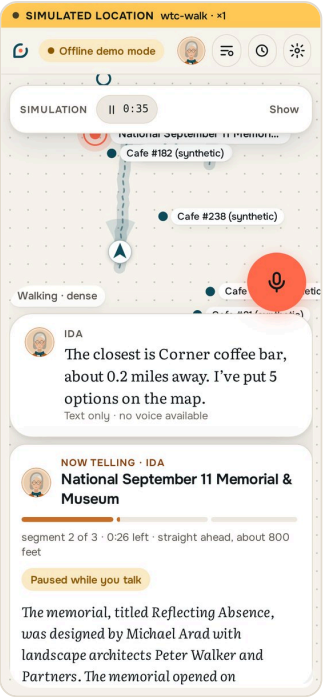
4 Approaching a target. First story has just started on the Lower Manhattan replay: spatial cue “straight ahead, about 800 feet”, target ringed on the schematic map, segment 1 of 3.



5 Active story. Segment 2 of 3 with per-segment progress ticks (these are the resume points). Pause, Skip and “Not that one” are the controls.



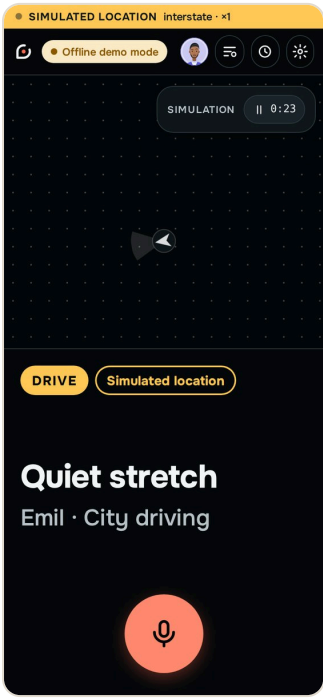
6 Interruption. Mic pressed mid-story: narration stopped, listening sheet open. Headless Chromium has no speech recogniser, so “Type instead” is offered; a typed question is captured in the packet.



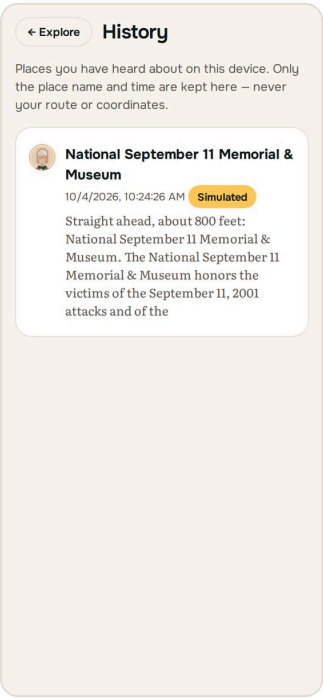
7 Nearby result on the map. “The closest is Corner coffee bar, about 0.2 miles away. I’ve put 5 options on the map.” Story card shows “Paused while you talk”. Places are synthetic.



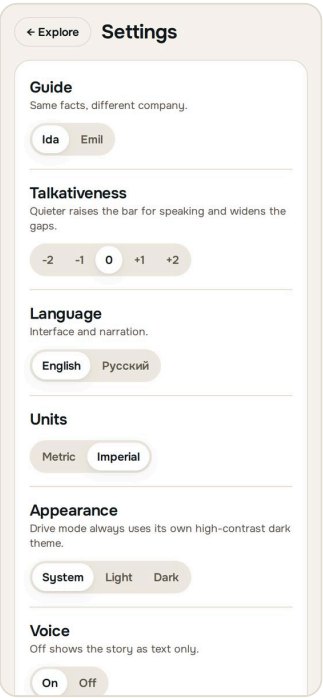
8 Resumed story. After the answer, the resume policy restarts the interrupted segment with a bridge (“Where was I? Ah yes, ...”) instead of restarting the story.



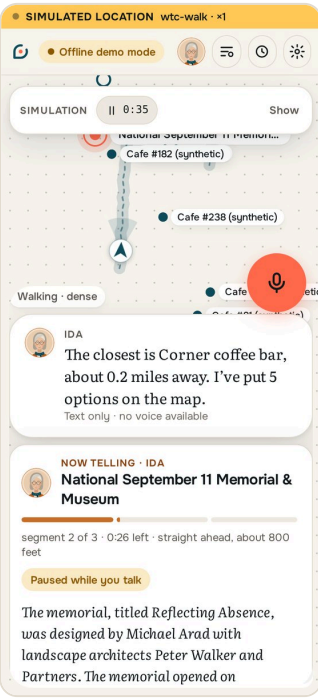
9 Drive-safe state. The highway regime on the synthetic I-40 replay switches on the DRIVE HUD by itself: dark palette, one status line, one large mic, no lists, no text input.



10 History. Places told in this session, kept in this browser only and labelled “Simulated”.



11 Settings. Guide, talkativeness, language (English or Russian), units, appearance, voice, connection mode (Auto, Live, Offline demo) and “Delete my data”.



12 Degraded delivery. With no voice available the answer arrives as text only and the card says so. The story continues. A second capture (packet) shows the app refusing to pretend it is live when no API is configured.

5.3 Desktop and operator surfaces

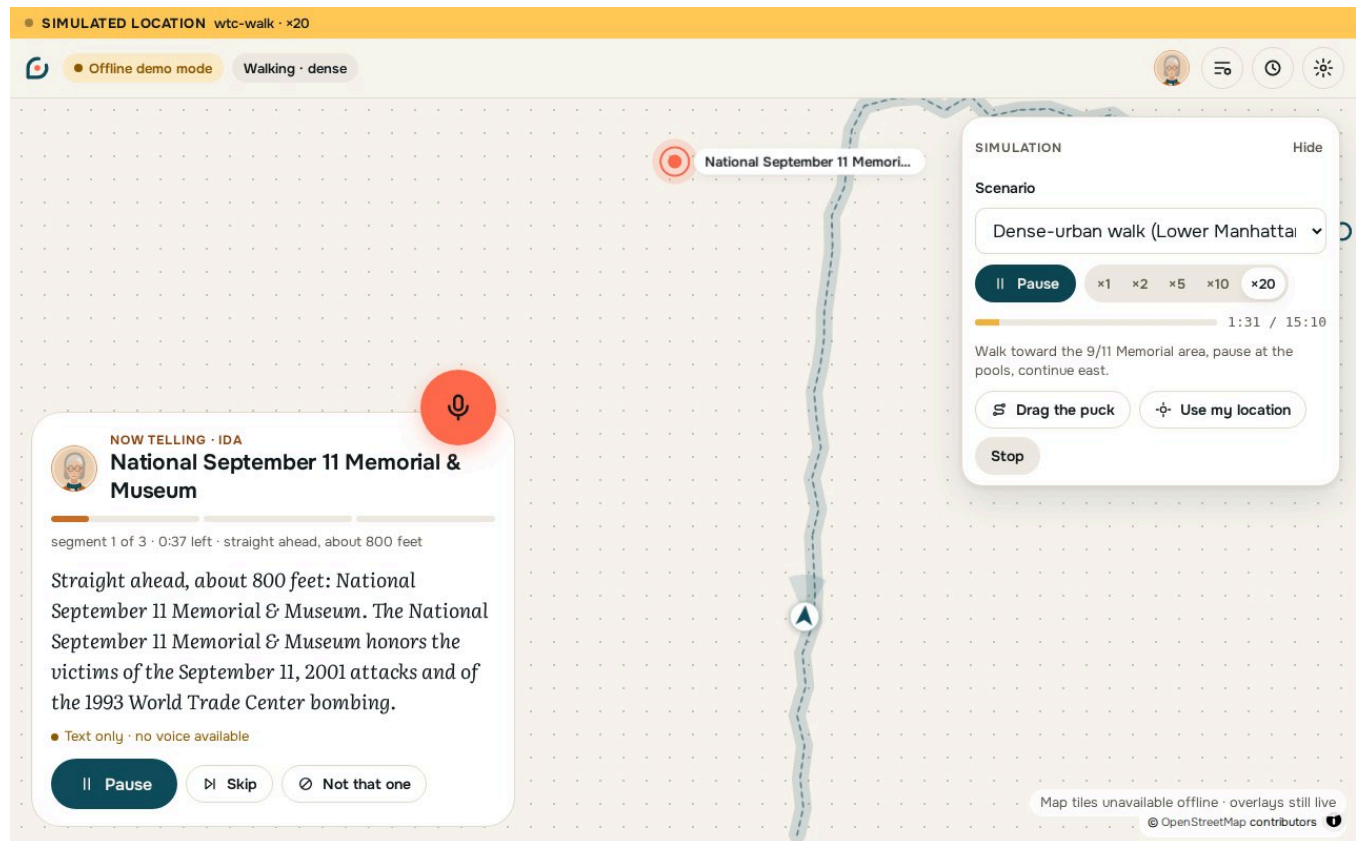


Figure 4. WebApp on desktop with the simulation panel (scenario, speed, progress). The same code that runs the product decides what to say.

Capture: 2026-10-04, offline demo, Lower Manhattan replay at 20x.

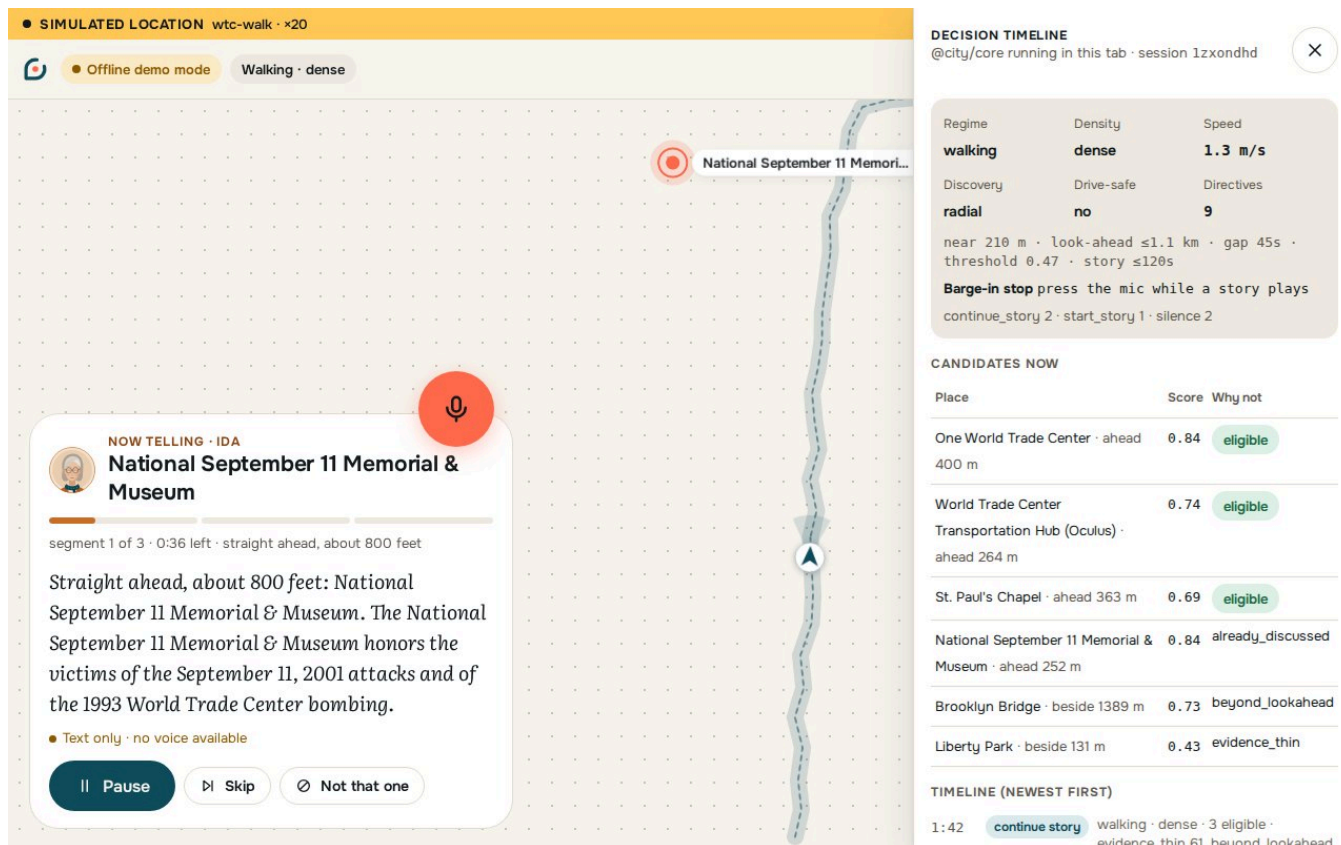


Figure 5. The explain drawer shows candidate scores and rejection reasons from the running decision code, the tool an operator would use to answer “why did it say that, or not?”

Capture: 2026-10-04, offline demo.

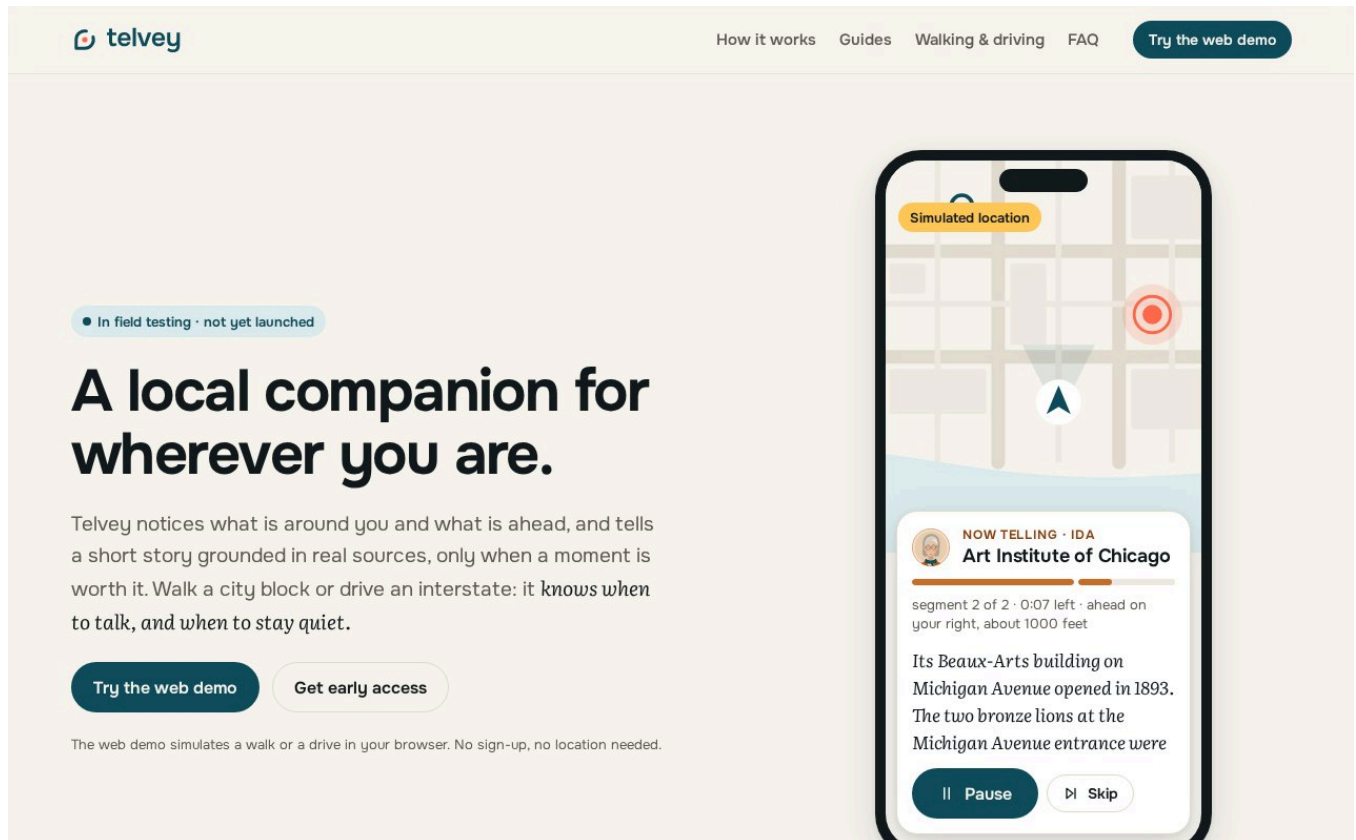


Figure 6. Public website, first screen, from the local production build. It is not deployed anywhere.

Capture: 2026-10-04.

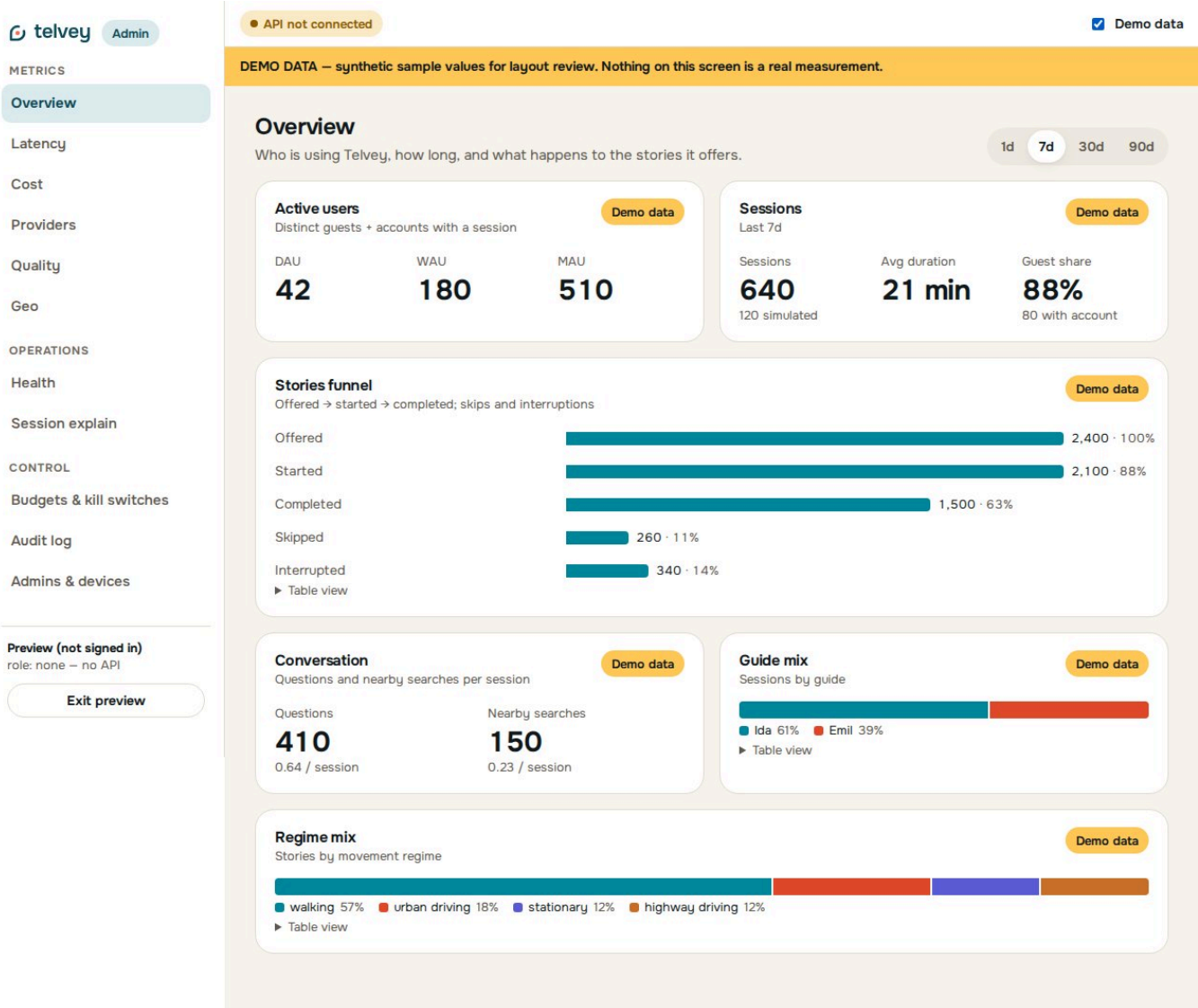


Figure 7. Web admin console in preview mode with the demo-data toggle on. The banner reads “DEMO DATA: synthetic sample values for layout review. Nothing on this screen is a real measurement.”

Capture: 2026-10-04. All numbers shown are illustrative.

The mobile admin surface was not captured. No device, emulator, Android SDK, macOS host or build credential exists in the environment, and the native app has never run. We chose not to render its screens through a web shim because that would be mistaken for a device capture. The screens exist in code (health, today’s sessions and stories funnel, latency, cost, recent errors, device pairing by QR or code) and are inventoried in `docs/MOBILE.md`; they are listed as not evidenced in section 17.

Demo recording. A 74-second screen recording of the offline demo (walk, mic press, typed coffee question, answer, resume, switch to the highway trip and the drive HUD) is in the comparison packet as `telvey-web-demo.mp4`. It carries the same limits as the screenshots.

5.4 What the product deliberately does not do

- It does not ask the driver anything. In driving regimes there are no tap-to-choose prompts, no Guide questions and no text input, and listening opens only on an explicit press.
- It does not tell stories about places it has too little evidence for. Thin evidence yields at most one short orientation line.
- It does not retell a place within 30 days of telling it to the same listener.
- It does not navigate. Directions are handed off to the user’s navigation app.

6. Brand and naming

Working name only. *Telvey* is a working brand. The screening below is a preliminary signal check made from live RDAP and DNS lookups and web, App Store and Google Play searches on 2026-09-27. It is **not** a trademark clearance, a freedom-to-operate opinion or legal advice, and no official trademark register was queried. The domains are **unregistered** as far as the last check showed; availability can change at any moment, and it was not re-verified on 2026-10-04 because the lookup tool was denied access in this session. A professional clearance search (Nice classes 9, 38, 39, 41 and 42; US, EU, UK and RU) is needed before any public use.

6.1 Candidates and the screen

About 900 names were generated and pre-screened by DNS; twelve were carried into a longer list and seven into full domain checks. The brief: short, pronounceable in English and Russian, verb-able, iconable, not tied to a city or to “tour”, and ideally a free .com or .ai.

Name	Idea	Domain result (2026-09-27 UTC)	Collision notes	Verdict
Telvey	<i>tell + convey / way</i>	.com, .ai, .app, .io, .dev, gettelvey.com free (RDAP 404); .co inconclusive	No product named Telvey found. Neighbours in other sectors: TelVue (broadcast), Telviva (telecom), Telva (magazine)	Selected (working); risk low to medium (“Tel-” is crowded in telecom, class 38)
Loreyon	<i>lore + yonder</i>	.com, .ai, .app free	Historic surname only; “LORE” marks exist	Cleanest alternative
Wendalong	<i>wend + along</i>	.com, .ai, .app free	Word-game apps	Long (9 letters)
Sayonder	<i>say + yonder</i>	.com, .ai, .app free	Sounds like “sayonara”	Weak brand
Tellaround	<i>tell + around</i>	.com, .ai, .app free	Close to “Showaround” (local-guide app)	Medium risk
Kolero	coined	.com registered 2025-08-11	Polish word, film, esports handle	Rejected
Hushway	<i>hush + way</i>	.com and .app registered	n/a	Rejected
Waytale	<i>way + tale</i>	.com and .app registered	Direct collision: “Waytale: AI Audio City Guide” is live on the App Store	Rejected

Table 1. Shortlist, abridged. Full longlist of twelve and every RDAP endpoint with its time window are in [docs/BRAND.md](#) and [benchmark/brand/domain_checks.json](#).

Why Telvey. It is not a dictionary word (room to become ownable), does not say “tour”, “city” or “AI”, has two clear syllables with the same pronunciation in English and Russian («Телвей»), and works as a verb (“just Telvey it”). A wake word “Hey Telvey” is a possibility only; a false-trigger evaluation has not been run and the app does not use a wake word. **Owner actions:** register telvey.com, telvey.ai and telvey.app, reserve handles, commission the clearance search. The Waytale result is also a reminder that the AI audio-guide space is moving; an AI audio city guide with selectable narrators is already live on the App Store.

6.2 Identity system



Figure 8. Brand board: mark, wordmark, palette, type, UI chips and Guide accents.

Source: `assets/brand/brand-board.svg` (original work; no prior brand, mascot or character is derived from).

The mark is an open speech bubble whose sharp lower-left corner doubles as a map-pin point; the opening faces upper right (“ahead”) and a single ember dot in the centre is the place being talked about. The idea is “a local who knows when to talk”: calm, warm, precise. Petrol (deep blue-green) is the ground, ember is reserved for the place and for listening, and silence is presented as a feature.

Role	Value	Note
Primary	Petrol 700 <code>#0F4C5C</code>	8.66:1 on paper (AAA)
Accent	Ember 500 <code>#FF6A4D</code> ; ember-ink <code>#B23A1E</code> for text	Ember is decorative only on light backgrounds (2.58:1); accent text uses ember-ink (5.44:1)
Drive theme	<code>#06090B</code> ground, text <code>#F4F7F8</code>	Every text pair at least 7:1 (AAA); nothing smaller than 28 px
Typography	Onest (UI, Latin and Cyrillic), Literata (story text), JetBrains Mono (admin numerals)	All SIL Open Font License
Contrast	Computed with the WCAG 2.2 relative-luminance formula	Table in <code>docs/BRAND.md</code>

Table 1. Palette and type, summary. Assets: mark, wordmark, horizontal lockups, iOS and Android adaptive icons, favicon and tokens are in `assets/brand/` and in the packet.

7. Original AI Guides

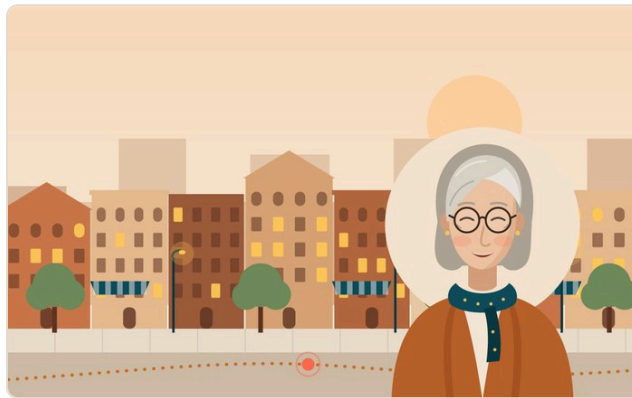
Two original Guides tell the same facts. They differ only in angle, length, humour, framing and voice. Neither is derived from an existing character, brand or person; portraits and scenes are hand-authored vector art with no real skyline, landmark or sign text.

	Ida	Emil
Concept	The unhurried storyteller who has time for every street	The sharp-eyed observer who gets to the point
Angle	People, origins, everyday life, links to earlier places	Hidden details, numbers, architecture, turning points
Verbosity (x regime budget)	1.15	0.80
Humour	0.30, gentle	0.65, dry, never snide
Opening and sign-off	Sensory scene, then the person; soft reflection	Hook detail first; crisp one-liner, then silence
Voice direction	Mature, warm, low-mid, rate 0.95	Clear, bright, mid, rate 1.05
Voice candidates (pending the D-010 listening test)	OpenAI sage (alt coral); Google Gemini Gacrux (alt Sulafat)	OpenAI ash (alt cedar); Google Gemini Iapetus (alt Sadachbia)

Table 1. Guide profiles. Source: `packages/core/src/guides/ida.json` , `emil.json` ; `docs/BRAND.md` section 3.



Ida, portrait



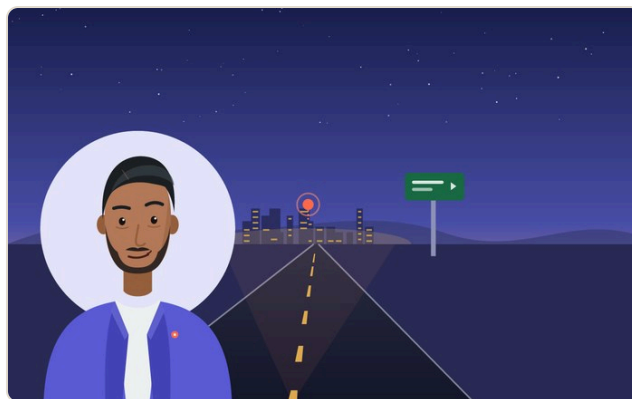
Ida, city-context scene (generic low-rise street, golden hour)



Ida, avatar crop



Emil, portrait



Emil, city-context scene (generic highway at dusk, unnamed town ahead)



Emil, avatar crop

How factual consistency is preserved. The Guide never chooses facts. A deterministic step builds a *StoryBrief* from the evidence pack (the approved facts, the angle, the length budget). Both Guides receive the identical facts at the same budget; the persona changes only how they are phrased. A language model writes the prose, and a *grounding check* then rejects any number, year or proper noun that is not in the approved evidence. If it fails, there is one constrained retry and then a deterministic template built from the same facts.

Check	Result	Label
Same lead fact for both Guides, across evidence packs	31 of 31	MEASURED-replay
Identical brief facts at the same budget	100% of packs	MEASURED-replay

Check	Result	Label
Grounding failures across all replay stories	0	MEASURED-replay (template prose)
Persona parameters differ (verbosity 1.15 vs 0.80, rate 0.95 vs 1.05, angle order)	Yes	MEASURED-tests
Pacing, structure and tone of live prose; human 1-to-5 rubric (naturalness, pronunciation, persona distinctness, factual preservation; English and Russian)	Not run	NOT RUN: needs a live model and raters

The honest limit: the deterministic fallback template is persona-neutral by design, so every Guide-difference claim at the *prose* level rests on a language model that has not yet been run. Acceptance B2 is therefore PARTIAL. The voice identifiers above are candidates that were seen in the providers’ public documentation on 2026-09-27; Russian quality and the fit of voice to persona must be judged by ear.

8. Architecture

PART III · HOW IT IS BUILT AND WHAT WAS MEASURED

The system has one organising idea: **a language model may write sentences, but deterministic code decides everything else and checks the model’s work**. Which place to talk about, when to speak, how long, whether it is safe, when to stay silent and how to resume are all computed by a pure, tested library. The model receives an already-decided brief and returns prose, which is then verified before anyone hears it. This keeps the behaviour that matters (timing, relevance, safety) testable without paying for a model, and it makes a model outage degrade to bland text instead of invented text.

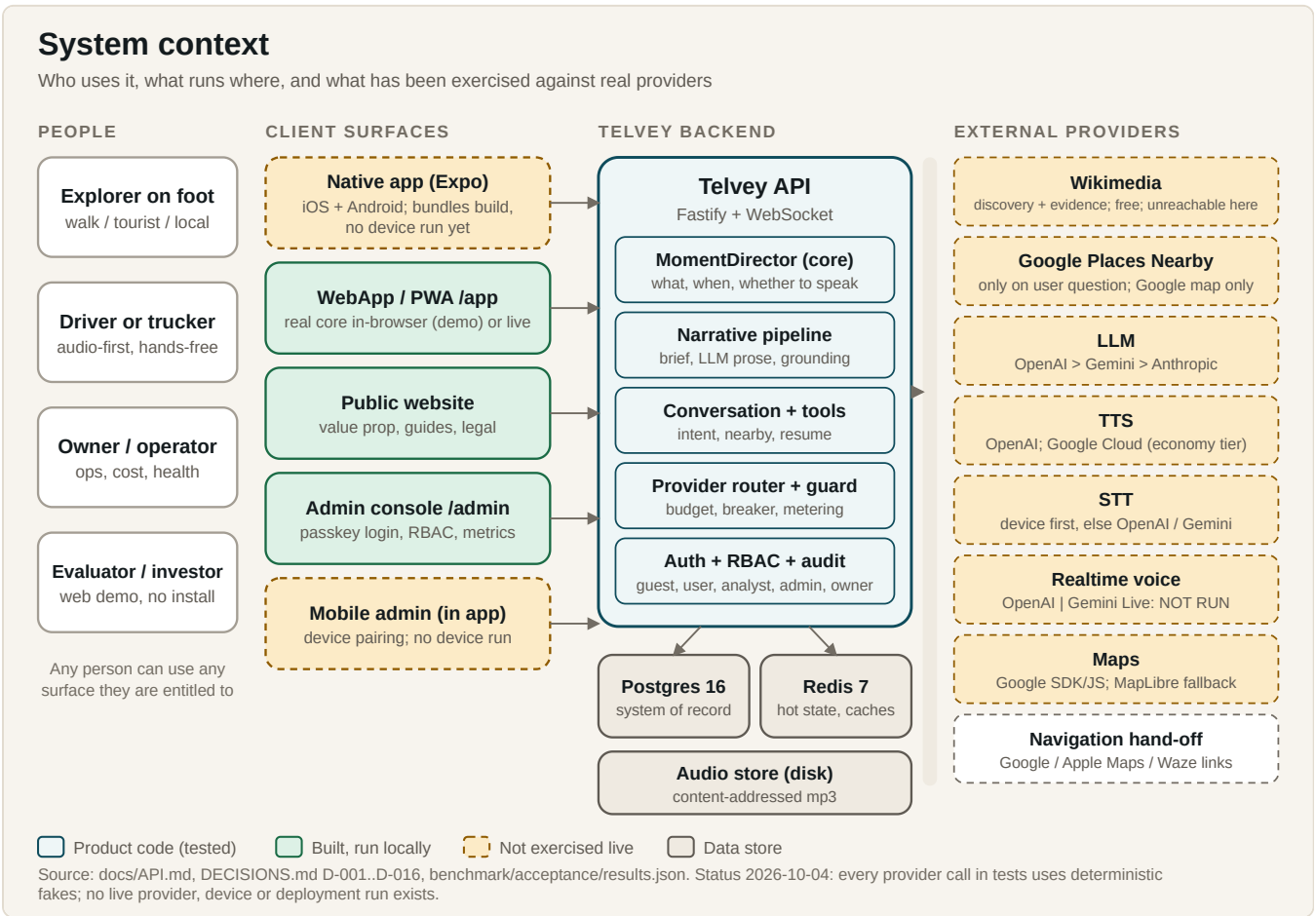
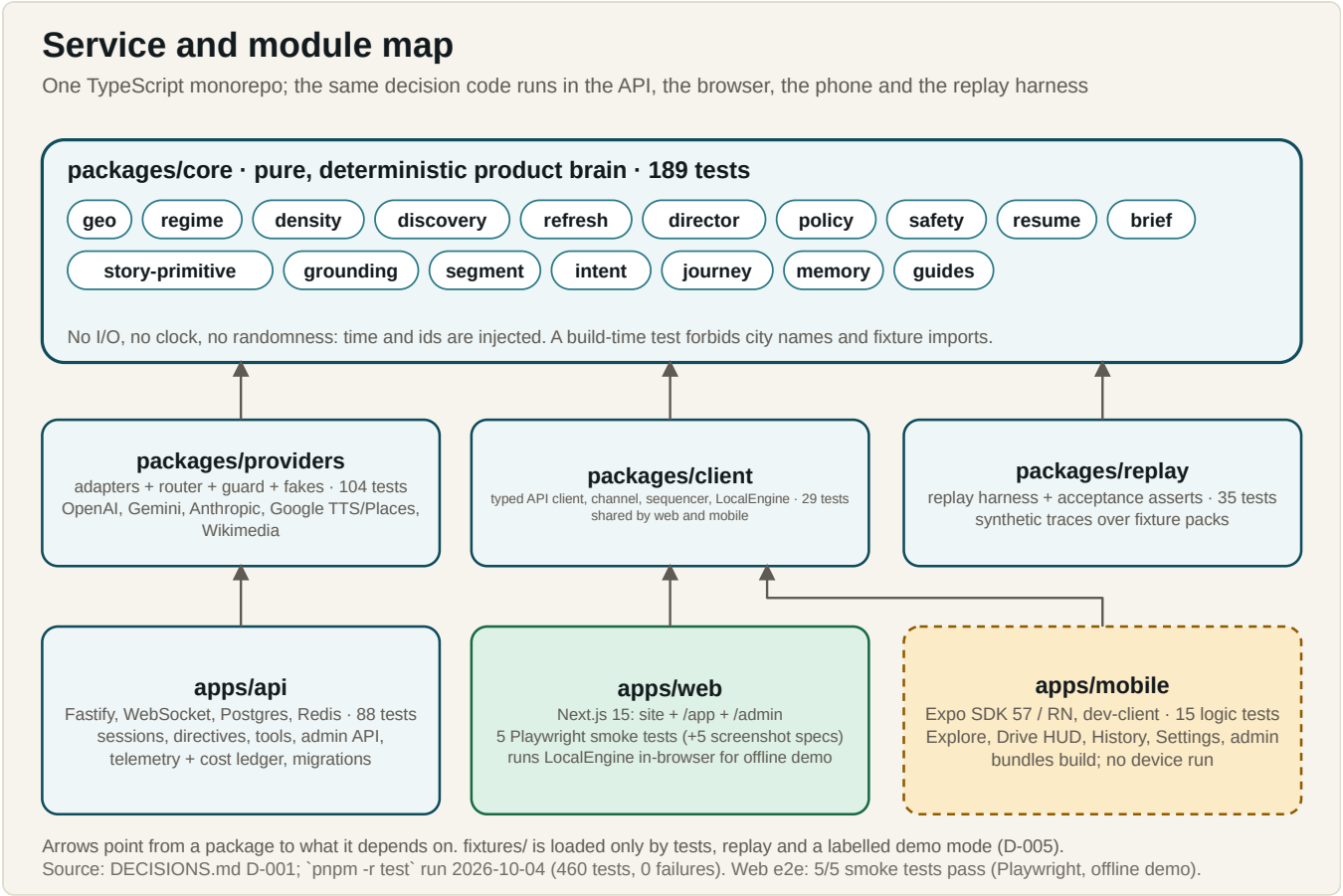


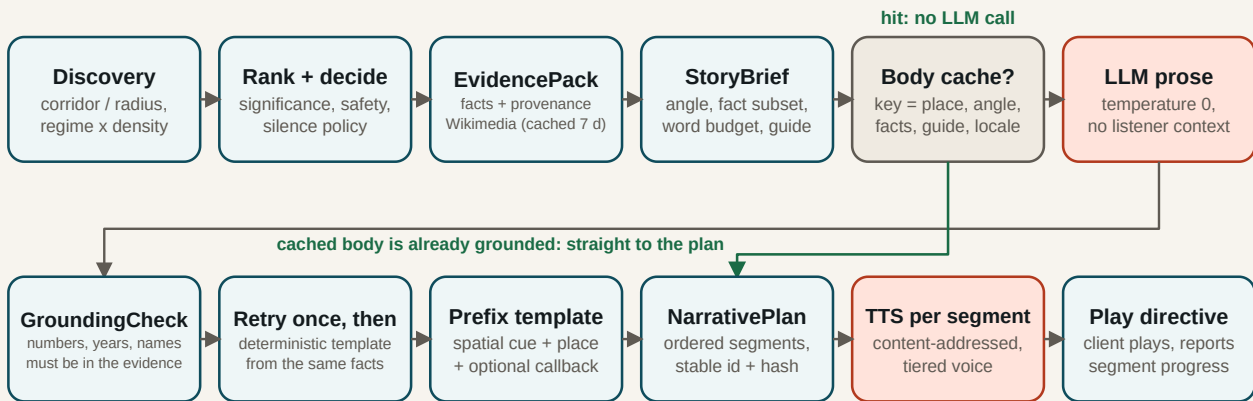
Figure 9. System context: people, client surfaces, the Telvey backend and external providers. Dashed boxes have never been exercised live.

Source: docs/API.md; DECISIONS.md D-001 to D-016; benchmark/acceptance/results.json.



Narrative pipeline: evidence to audio

The model writes sentences; deterministic code decides everything else and checks the model's work



Why this shape matters commercially

1. The cacheable unit is the context-free story body: one LLM + TTS spend can serve many listeners (production hit rate NOT YET MEASURED).
2. A model outage degrades to template text for the same target; the model never picks the place, the timing or the length.
3. Every spoken sentence is checked against evidence before it is voiced, so the failure mode is bland, not invented.

 Deterministic code Generative model (paid) Cache / store

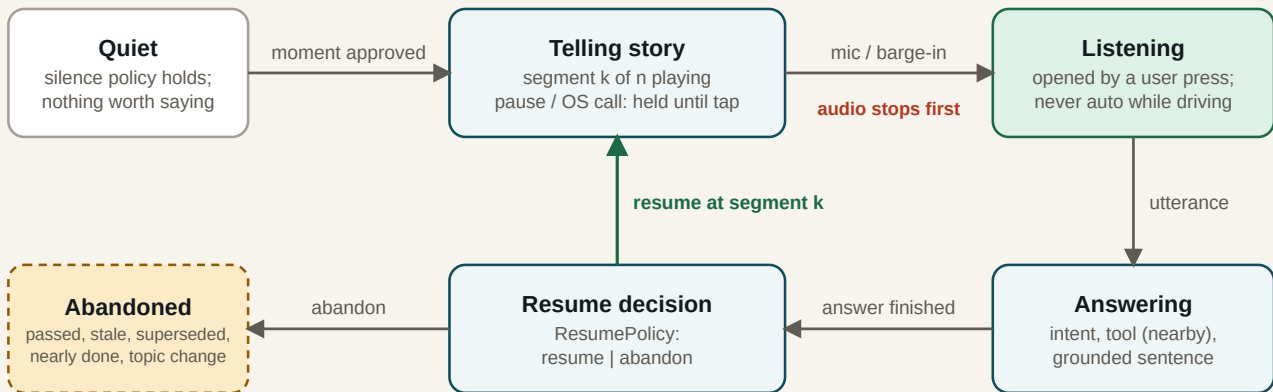
Source: DECISIONS.md D-003, D-009, D-018; packages/core (brief, story-primitive, grounding, segment). TTS is a generative provider call even though text is fixed.

Figure 11. Narrative pipeline from discovery to audio. Everything except the model call and the voice call is deterministic code.

Source: DECISIONS.md D-003, D-009, D-018.

Conversation, interruption and resume

Barge-in stops audio first; the resume decision is a deterministic policy, not a model call



Deterministic rules behind the arrows (packages/core/src/resume.ts, safety.ts)

Resume restarts at the START of the interrupted segment (a sentence group); a bridge phrase is added if the pause exceeded 8 s or an answer was spoken.
 Abandon when: a newer plan exists; the target is passed (driving: more than 50 m behind; walking: more than 400 m away); the pause is stale (walking 3 min, driving 90 s); or 85% of the segments were already told. An explicit topic change never auto-resumes the old story (acceptance C3).
 If listening closes with no utterance, the same policy decides on the next tick.

 Server policy state
 User-initiated
 Terminal

Source: DECISIONS.md D-008/D-009; acceptance C1, C3, E3. Exercised in tests with fake providers; not yet on a physical device.

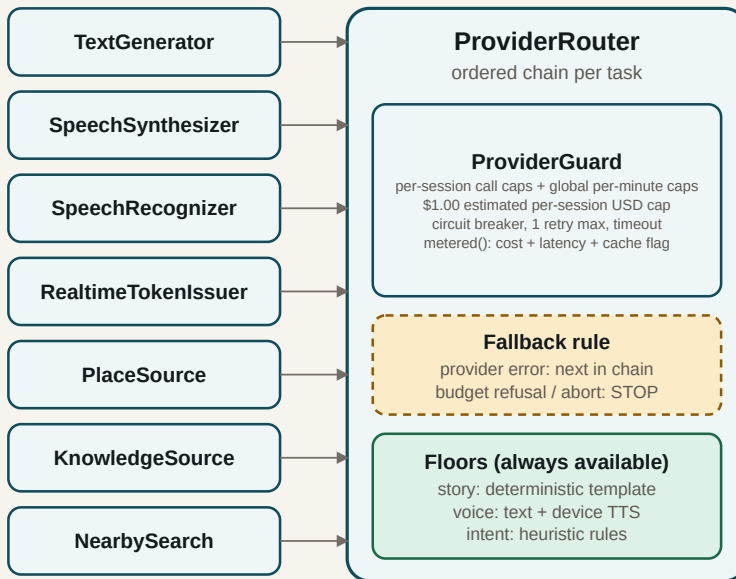
Figure 12. Conversation, interruption and resume as a state machine. Audio stops first; the resume decision is a policy, not a model call.

Source: DECISIONS.md D-008, D-009; acceptance C1, C3, E3; exercised with fake providers, not on a device.

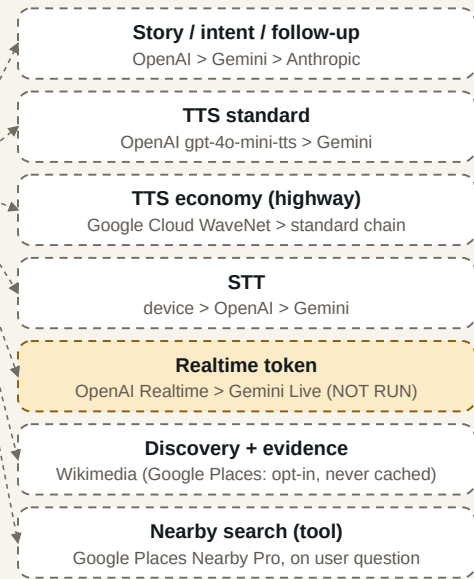
Provider abstraction, routing and budget

Every paid or external call passes one guard; a refusal stops the chain instead of fanning out

CORE INTERFACES



ADAPTERS (chain order)



Deterministic fakes implement every interface, with injectable simulated latency. All 460 tests and the whole benchmark run on fakes: no real provider has been exercised.

Model ids are env-configurable; ``pnpm bench:providers`` checks ids against each provider's model list before any switch (NOT YET RUN: no keys).

Source: packages/providers (router.ts, resilience.ts, metered.ts, pricing.ts), docs/PERFORMANCE_COST.md §12.

Figure 13. Provider abstraction. Every paid or external call passes one guard that enforces budgets, a circuit breaker and at most one retry; a budget refusal stops the chain rather than fanning out to the next provider.

Source: packages/providers; docs/PERFORMANCE_COST.md section 12.

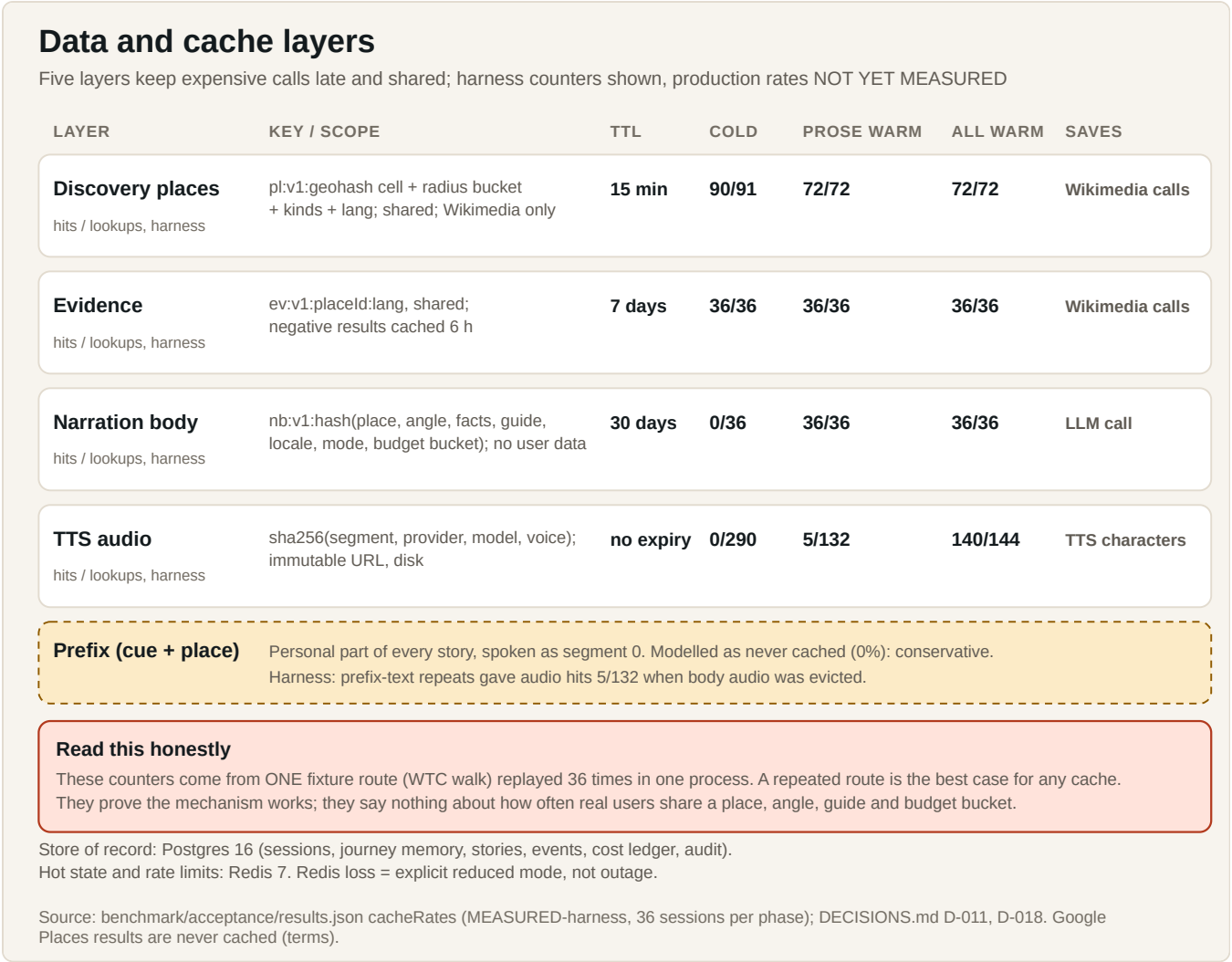
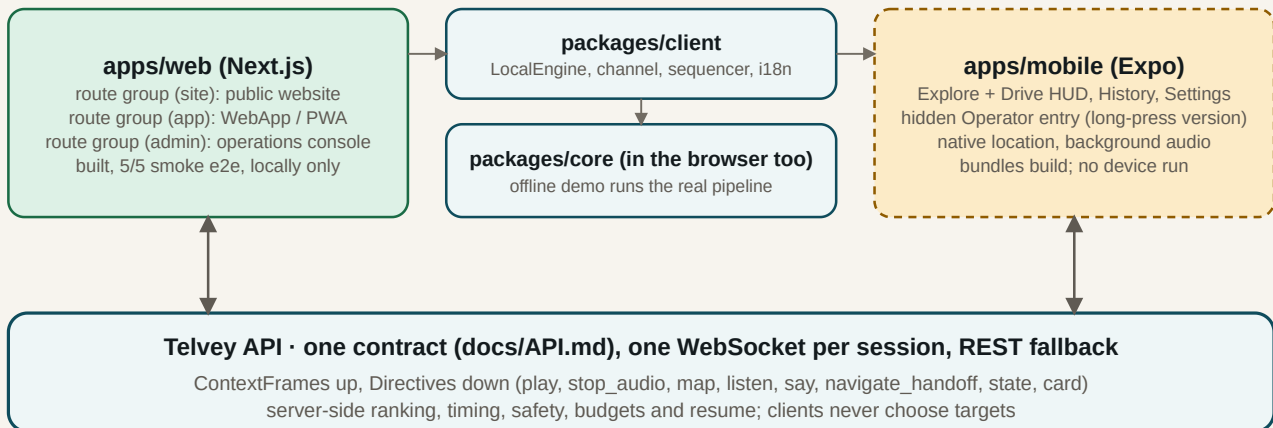


Figure 14. Data and cache layers, with the harness counters per phase.
Data: benchmark/acceptance/results.json cacheRates (MEASURED-harness: one fixture route replayed 36 times in one process; not production hit rates).

Surfaces: web, mobile, admin

One backend and one decision core; each surface is a sensor, a renderer and an audio player



Admin access: web console uses WebAuthn passkeys (first owner bootstrapped by a one-time server CLI code). A signed-in admin issues a 5-minute pairing QR; the phone redeems it for a revocable, role-scoped device token kept in secure storage. No static password or secret ships in any client. Every admin action is audited.

Source: docs/API.md auth model, docs/MOBILE.md §1, DECISIONS.md D-002, D-013. Mobile admin has not been run on a device.

Figure 15. Web, mobile and admin surfaces share one API contract and one decision core.

Source: docs/API.md; docs/MOBILE.md section 1; DECISIONS.md D-002, D-013.

8.1 Major choices and what they cost

Choice	Why	Trade-off accepted
Server owns product authority (D-002). Clients are sensors, renderers and players	One policy for every surface; clients cannot fork behaviour	Offline behaviour needs a bounded prefetch and a local copy of the engine (built for demo mode)
LLM only writes or interprets (D-003) with a deterministic grounding check and template floor	Failure mode is bland, not invented; target choice cannot be corrupted by a generator	Prose quality ceiling is the template floor when the model fails; live grounding pass rate is unmeasured
Location-first, no city coupling (D-004, D-005)	The owner's global requirement; a build-time test forbids city names and fixture imports in production code	Real candidate density is untested because live discovery was unreachable
Regimes and trajectory corridors (D-006, D-007)	Highway reasoning needs “what is ahead”, not “what is near”; silence is a first-class outcome	Thresholds are hand-set and only checked on synthetic traces
Exact resume (D-009)	Interrupting must not lose the story	Resume restarts the segment, not the sentence
Hybrid voice, not always-on realtime (D-010, provisional)	Cost is bounded to spoken minutes; no open microphone while driving	Turn-taking is likely slower than realtime; unmeasured
Shared context-free story body + uncached personal prefix (D-018)	One model call and one voice render can serve many listeners; no personal data in the shared cache	Repeat listeners hear one canonical telling; a second short voice call per story; seam audible if voices differ
Tiered voice (D-019)	Highway narration on a roughly 4x cheaper voice	Economy voices are unauditioned candidates; city stays on the standard voice
Postgres plus Redis, single VPS (D-011, D-015)	Cheap, simple, replaceable	Sticky WebSockets, local audio and process-local budget windows block multi-instance; see section 16
Privacy-minimal telemetry (D-012)	Coarse geohash only, no raw traces unless the user opts in	Limits some product analytics

9. Technology and provider decisions

No provider decision rests on a live measurement. No provider key was supplied, so every row below is a design or list-price judgement. The decision procedure for replacing it with evidence is in section 11.

Layer	Choice (default)	Alternatives considered	Reason (and its evidence level)	Lock-in risk	Fallback or migration
Language	TypeScript strict, Node 22, npm monorepo	Separate server and mobile stacks	One type system across every boundary; replay harness runs the production code. MEASURED-tests	Low	n/a
Story and intent model	gpt-6-luna ; chain OpenAI, Gemini, Anthropic	Gemini Flash-Lite, Claude Haiku or Sonnet	Lowest list price; the model is under 2% of session cost, so choose on grounding rate and latency, both unmeasured . ESTIMATED	Low: interface plus three adapters	Template story floor; kill switch
Voice (TTS)	gpt-4o-mini-tts standard; Google Cloud WaveNet economy tier for highway (in this report “economy tier” always means Google WaveNet)	tts-1, Gemini TTS, ElevenLabs, Deepgram, Cartesia	The single biggest cost line; WaveNet is about 4x cheaper per character. Voice quality unaudited . ESTIMATED	Medium: per-persona voice ids	Standard chain, then device voice and text
Speech input	Device recognition first, else gpt-4o-mini-transcribe	Gemini transcribe, Deepgram, ElevenLabs Scribe	Free on device; under \$0.001 per question otherwise. ESTIMATED	Low	Server STT, then typed text
Realtime voice	None selected; hybrid loop runs	OpenAI Realtime, Gemini Live, GPT-Live-1	Cost and risk reasons only (section 11). NOT RUN	Medium	Hybrid loop
Discovery and evidence	Wikimedia (Wikipedia and Wikidata): free, cacheable, poolable	Google Places, OSM Overpass	Free, shareable across users, strongest on significance and history; rate-limited. Unreachable from the build sandbox, so live discovery is untested.	Low	Self-hosted Wikidata and OSM extract; fixtures for demo only
Nearby search (tool)	Google Places Nearby Search Pro, only on a user question	Wikimedia, Overpass, Mapbox Search	Needs business data (open now, category). \$0.032 per call. ESTIMATED	Medium	Spoken “limited” phrase; alternative source
Map	MapLibre with OpenStreetMap on web by default; Google Maps JS when a key is set; Google Maps SDK on Android; Apple Maps on iOS without a Google key	Mapbox, Google everywhere	Native Maps SDK is free; see the terms conflict below	Medium	Schematic overlay when tiles fail
Web and mobile	Next.js 15 (site, app, admin in one app); Expo SDK 57 with React Native 0.86	Separate sites; Flutter	One design system, one deployment; native modules need a dev client. Bundles compile; no native build ever ran. MEASURED-tests	Low	n/a
Data stores	Postgres 16 and Redis 7	Managed services, SQLite	Standard and cheap; Redis loss is an explicit reduced mode. MEASURED-tests	Low	Managed services when scaling
Admin auth	WebAuthn passkeys; device-pairing for the phone	Passwords, email OTP	No static secret exists anywhere to leak. MEASURED-tests	Low	Invite codes
Hosting	One VPS, Docker Compose, Caddy TLS, GitHub Actions	Managed PaaS	Cheapest MVP path; never built on a real host	Low	Any container host

Google Places terms shape the architecture, and one rule is not yet enforced. Google’s service-specific terms (retrieved 2026-09-27) allow only latitude and longitude from Places to be cached, for at most 30 consecutive days; place identifiers are exempt; and Places content must not be shown on a non-Google map. The code honours the first two: Places results are never put in the shared discovery cache, discovery uses Wikimedia, and Places is called only for an explicit question. The third is documented but **not enforced in code**: the web map defaults to OpenStreetMap, and on iOS the app uses Apple Maps unless a Google Maps SDK key is supplied, so a live Places result could be drawn on a non-Google map. This must be closed (a Google map wherever Places content appears, or a different nearby-search source) before any live Places traffic. Legal should also review whether generated narration derived from Places fields may be cached.

Further terms and licences, checked 2026-10-04. (1) Google’s terms also contain a Places UI Kit exception that allows a non-Google map, and it prevails over the non-Google-map rule; whether text-to-speech audio derived from Places fields counts as Google Maps content is an open legal question. (2) Wikipedia text is licensed CC BY-SA 4.0 (Wikidata is CC0). Narration adapted from it needs attribution and may carry ShareAlike obligations that reach cached story text and shared audio; no attribution is shown to the listener today and legal review is required. (3) The default web map uses OpenStreetMap raster tiles; the public tile servers are not for production traffic, so a production tile source and attribution are needed. (4) Google’s current pricing page lists Standard and WaveNet voices at \$4 per million characters but under “Legacy TTS models”; the current-generation Chirp 3 HD voice is \$30 per million, about 1.8 times the standard voice used here. If a current-generation voice is required, the economy tier’s cost advantage largely disappears, so the driver plan’s viability rests on a legacy voice family.

Model-name conflict. The OpenAI pricing page on openai.com and the developer documentation listed different flagship model names on 2026-09-27 (“GPT-5.6 Sol/Terra/Luna” versus “GPT-6 Astra/Sol/Luna”). The developer docs and the individual model pages agreed with each other, so the model identifiers and prices used here come from them. The identifiers must be confirmed with the provider’s model-list call when keys arrive; the provider benchmark script does exactly that and has so far run with no keys.

10. Performance results

10.1 What was measured, and on what

All latency figures below are **MEASURED-harness**: the real API (Fastify, Postgres 16, Redis 7, WebSocket) runs in-process on loopback in a sandbox with two CPUs, and every external provider is a deterministic fake wrapped in a seeded log-normal delay chosen to resemble that class of API. The delay profile is an assumption, not a measurement of any provider: story LLM 1,100 ms median and 2,600 ms 95th percentile; follow-up LLM 700 and 1,600 ms; text-to-speech 250 ms plus 2.5 ms per character; speech-to-text 450 and 1,100 ms; nearby search 250 and 650 ms. The harness therefore proves the orchestration (what runs in parallel, that stale turns are dropped, that nothing waits needlessly). It does not prove real provider speed, tail behaviour or network cost.

Three further limits matter. Fake prose is shorter than real model prose, so the harness voice terms are optimistic. Loopback hides network time (add roughly two round trips; an **ASSUMED** 40 to 120 ms each on LTE, not measured). And each figure is 36 seeded sessions after one excluded warm-up, so a margin of a few percent against a target is within noise.

10.2 Latency against targets

The template contains no numeric targets, so these targets were proposed by the build agents and need owner sign-off.

Interaction	Target p50 / p95	Result p50 / p95	Verdict
Approved moment to first audio, nothing cached	2.5 s / 4.5 s	1,663 / 2,683 ms	Met in simulation
Cached story to first audio, body cached, audio evicted	0.5 s / 1.0 s	448 / 883 ms	Met in simulation (conservative case)
Cached story to first audio, fully warm	0.5 s / 1.0 s	6 / 28 ms	Met on loopback; a cache read only
Speech end to answer audio, “coffee nearby”	2.0 s / 3.5 s	1,250 / 2,328 ms	Met in simulation
Speech end to answer audio, contextual follow-up	2.0 s / 3.5 s	1,950 / 3,380 ms	Met by a 3% margin at the 95th percentile; was 2,840 / 4,915 ms (missed) before streaming
Speech end to nearby results on the map	1.5 s / 2.5 s	701 / 1,545 ms	Met in simulation
Interrupt to stop signal (server round trip)	150 ms / 300 ms	1.0 / 1.5 ms	Met on loopback; press-to-silence on a device not measured
Context frame ingest	n/a	0.1 / 0.2 ms	Local compute is negligible
Native app launch to usable Explore	1.5 s / 3.0 s	none	NOT RUN : no build exists
WebApp first usable state	1.5 s / 3.0 s	none	NOT YET MEASURED : only a functional end-to-end pass exists

Table 1. All rows MEASURED-harness with simulated providers unless the result says none. Source: `benchmark/acceptance/latency.json`, n = 36 per metric.

The follow-up row is the most fragile number in the report. It depends on an assumption that the model’s first word arrives after 45% of its simulated total latency and that voice rendering starts per sentence. With real providers it may move either way. Treat it as “inside target in simulation by a thin margin”, not as a guarantee.

Hybrid voice path vs proposed targets: all p95 inside targets

MEASURED in-process harness (real API + Postgres + Redis + WebSocket, loopback) with SIMULATED provider latency (seeded log-normal profile); excludes mobile-network RTT and device audio start. n = 36 per metric. Run 2026-10-04.

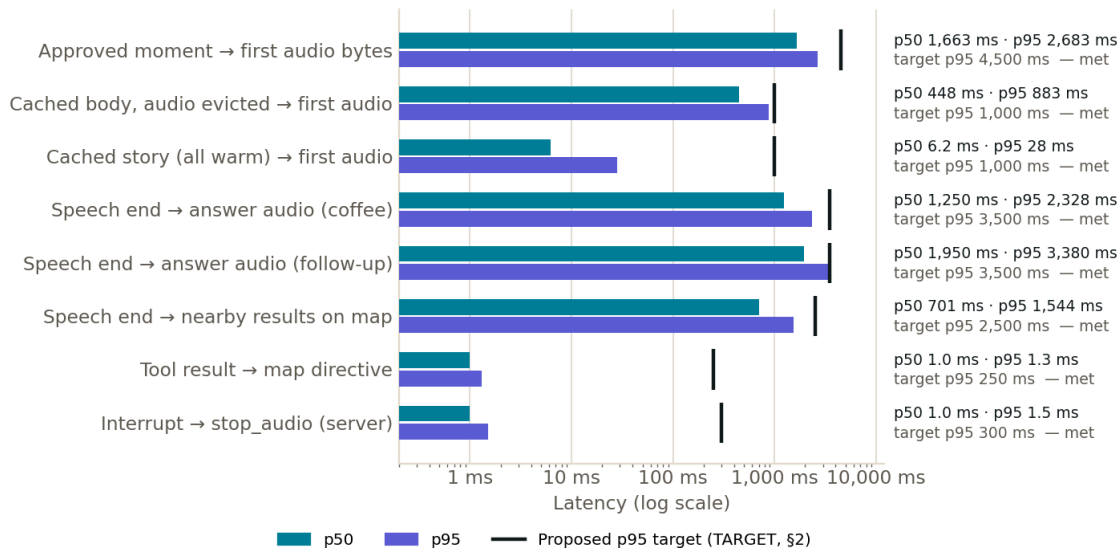


Figure 16. Latency percentiles against targets (simulated providers, loopback).

Data: `benchmark/acceptance/latency.json`. MEASURED-harness.

Slowest stages. Provider time dominates everything: the story language model (about 1.1 s simulated) and voice rendering of segment 0, then speech-to-text for questions. Our own overhead is tiny (frame ingest 0.1 ms; stop signal about 1 ms; audio file serving 3 ms). The cheapest latency win is the story cache: when a story body is already cached, no model is called and first audio is the short prefix, which cut the cached case from 1.06 s to 0.45 s median in the harness.

10.3 Silence is overstated for walking, and slightly for driving

The replay harness reports how much of a trace was silent, and the headline numbers were 84% to 91% silent on the city walks and 98.7% on the 105-minute I-40 trace. Those numbers are real outputs of the harness but they are **not the silence a listener would experience**, because the replay measures speech duration from the *template* story text, which is much shorter than a real model’s prose.

The cost model uses real prompt budgets (stories filled to about 85% of the word budget at roughly 2.4 spoken words per second) and gets a different picture:

Situation	Replay silence (template-length speech)	Narrated share with model-length stories	Derived silence	Label
Dense city walk, 30 min	84% to 91%	20.1 of 30 minutes (about 67%)	about 33%	DERIVED from the cost model
Urban drive, 30 min	91% (Golden Gate trace, mixed driving and walking)	7.2 of 30 minutes (24%)	about 76%	DERIVED
I-40 highway, 30 min	98.7%	1.1 of 30 minutes (3.6%)	about 96%	DERIVED

Table 1. The replay figures are MEASURED-replay; the narrated minutes come from `benchmark/cost/cost_model.json`; “derived silence” is one minus the narrated share.

So the honest claims are: on a sparse highway the product is silent roughly 96% of the time (longest silent gap about 32 minutes in the replay), which supports the design; but on a dense walk the current cadence has the Guide talking about two thirds of the time. That is a product-policy choice (story length and minimum gap), it drives cost, and no listener has judged whether it is too much. The owner may

want a quieter default for walking.

10.4 Cache behaviour and call discipline

The cache table is in [Figure 14](#). The cold phase shows no story-body hits and no audio hits (0 of 36 and 0 of 290); with prose cached but audio evicted, body hits are 36 of 36 and first audio is 448 ms; fully warm, audio hits are 140 of 144. These are counters from one fixture route replayed 36 times in one process, which is the best case for any cache. **Production hit rates are NOT YET MEASURED** ; the admin console exposes the counters so they can be read as soon as there is traffic.

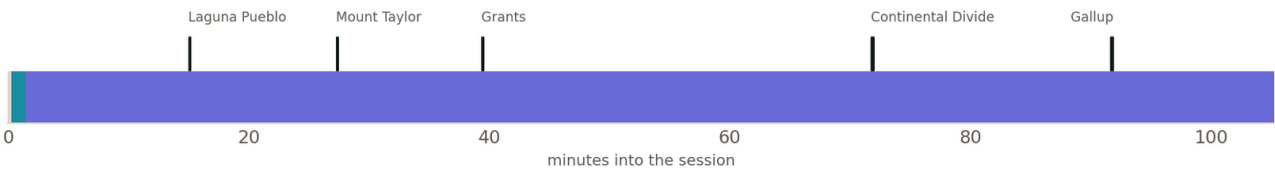
Call discipline against paid or rate-limited providers is the other measured strength (**MEASURED-replay** and **MEASURED-tests**):

- Automatic discovery makes 18 to 77 place queries per hour against 3,600 location fixes per hour; the I-40 replay makes 18.2 per hour.
- An empty discovery result makes 4 place-source calls in 300 frames, not 300.
- A provider timeout makes 3 attempts in 300 frames (one retry plus a circuit breaker); a quota error (429) or an invalid credential (401) makes exactly 1.
- If the language model keeps producing unsupported text, the same target is told from a template with at most 2 model calls per story; if the voice service is down, text is delivered and at most 4 voice attempts are made.

Silence is the normal state on the highway; the regime adapts without a restart

MEASURED-replay: synthetic traces over fixture POI/evidence packs through the production core (benchmark/replay, 2026-09-27).
Band = movement regime; dark bars = narration at template length (LLM stories fill up to the regime budget).

I-40 westbound, heading corridor (105 min): 5 stories, silence 98.7%, longest gap 32 min, 18 place queries/h



Highway → outskirts → downtown → park → walk (38 min): 8 stories, silence 93.5%, longest gap 8 min, 48 place queries/h

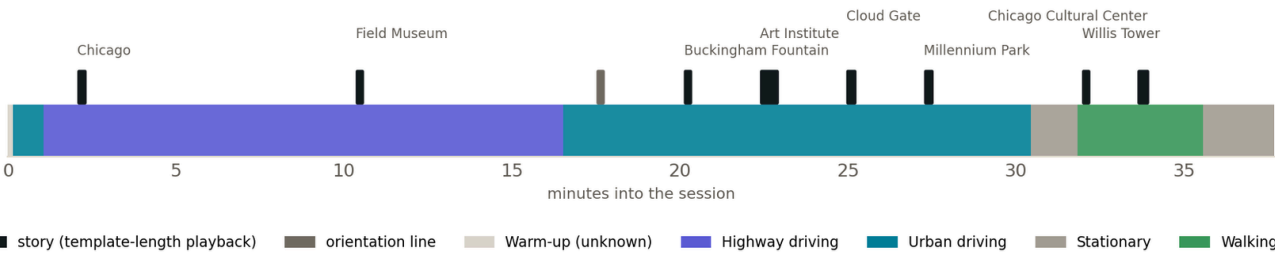


Figure 17. Replay timeline across the highway-to-downtown transition: regime, density, look-ahead and story starts.
Data: benchmark/replay/transition.json (synthetic trace; MEASURED-replay).

11. Realtime voice benchmark

NOT RUN: credentials not supplied. No OpenAI or Google key exists in this environment, so no realtime session was opened. Every measured cell below is empty on purpose. The table shows what the benchmark will record, what is known without a key, and how the decision will be made.

Metric	A: OpenAI gpt-realtime-2.1-mini	B: Google gemini-3.8-live	Selected / fallback rationale
Connect p50 / p95	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Pending; token issuers are implemented for both
Speech end to final turn p50 / p95	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Pending
Speech end to first audio p50 / p95	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Hybrid loop in simulation: 1,250 / 2,328 ms (coffee), 1,950 / 3,380 ms (follow-up), MEASURED-harness
Barge-in stop p50 / p95	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Hybrid server round trip 1.0 / 1.5 ms on loopback; device not measured
Tool reliability	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Fake tool results: invalid rows dropped 36 of 36
Factual preservation	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Pending; grounding applies to the hybrid path
Road-noise handling (false starts, missed turns)	NOT RUN: needs credentials and a noise fixture	NOT RUN	Pending
English quality (1 to 5)	NOT RUN: human rubric	NOT RUN: human rubric	Pending
Russian quality (1 to 5)	NOT RUN: human rubric	NOT RUN: human rubric	Pending; below 3 of 5 triggers re-evaluation
Guide A persona (Ida)	NOT RUN: human rubric	NOT RUN: human rubric	Pending
Guide B persona (Emil)	NOT RUN: human rubric	NOT RUN: human rubric	Pending
Reconnect success	NOT RUN: credentials not supplied	NOT RUN: credentials not supplied	Hybrid-path reconnect passes (exactly-once resume), MEASURED-tests
Active-minute cost	\$0.0117	\$0.0087	ESTIMATED list price at 30% listener and 40% assistant audio; the larger gpt-realtime-2.1 is \$0.0369
Client complexity	Token issuer implemented (WebRTC or socket)	Ephemeral-token issuer implemented	Both implemented server-side; no client has opened a session

Decision status. None selected. D-010 stays provisional. The hybrid loop (speech-to-text, rules or model, grounded answer, text-to-speech) runs today and is the default because it works without realtime keys, bounds cost to spoken minutes and avoids an open microphone while driving. Always-on realtime was rejected on cost (it grows with open-session time, and on OpenAI each turn re-bills the history) and on risk. OpenAI's full-duplex GPT-Live-1 at \$0.05 per session minute was rejected because it bills during silence, which is the product's most common state.

How the decision will be made. (1) Put keys in the environment, never in the repository. (2) Run the provider benchmark, which first verifies model identifiers against each provider's model list (the documentation disagreed on names), then measures story latency, tokens, cost and grounding pass rate on a fixed corpus (both Guides, English and Russian), voice time-to-first-byte and cost per minute, and speech-to-text latency. (3) Run the six realtime scripts (story interruption with coffee, contextual follow-up, barge-in "No, I meant parking", road noise, Guides and languages, session lifecycle) on each candidate with the same utterances as the conversation tests and one recorded road-noise fixture; two raters score the 1-to-5 rubric. (4) Choose by, in order: grounding pass rate of at least 95% after one retry; meeting the latency targets; lowest cost per narrated minute among those that pass; persona distinctness of at least 3.5 of 5. Re-evaluate if a price changes by 25% (the Gemini promotional prices double on 2027-01-01), grounding falls below 90% over seven days, 95th-percentile latency regresses past its hard threshold, or Russian quality is below 3 of 5.

12. Cost model

12.1 Basis and dates

Every dollar figure in this report is **ESTIMATED** : a list-price model, not a bill and not telemetry. Provider prices were read from official pricing pages on **2026-09-27** (the provider benchmark never ran, so no price has been checked against a console). Google's promotional Gemini prices double on 2027-01-01 and are modelled at the 2027 price. The Google Cloud text-to-speech price (\$4 per million characters for Standard and WaveNet) is the existing 2026-09-27 row; the official page could not be re-fetched on 2026-10-04. The Azure speech price is third-party and unverified, and is not used. Google's monthly free quotas (for example 5,000 free Nearby Search calls) are ignored, so the model is conservative at pilot scale.

What is measured and what is assumed: **call counts** come from deterministic replays of the production code (stories, place queries and evidence lookups per hour) and **token counts** from the real prompt builders; everything else is an explicit assumption: stories are filled to 85% of their word budget, 10% of stories need a grounding retry, 6.0 characters per spoken word, one follow-up question in a 30-minute walk, a one-cent-per-gigabyte egress price. Usage mixes, sessions per user and conversion rates are hypotheses.

12.2 Per-session cost

Default configuration (gpt-6-luna for text, gpt-4o-mini-tts for voice, gpt-4o-mini-transcribe, Wikimedia discovery, Google Nearby Search only when asked), no shared story cache. All figures in US dollars, **ESTIMATED** .

Component	30-min walk	30-min urban drive	30-min highway drive	60-min interactive walk	30-min walk + 5 realtime min	High-density worst case
Map loads, routes, details	0	0	0	0	0	0
Automatic discovery (Wikimedia, free)	0	0	0	0	0	0
Nearby search (when asked)	0	0	0	0.032	0	0.096
Story and answer text (LLM)	0.0027	0.0013	0.0002	0.0057	0.0027	0.0035
Voice (TTS)	0.3034	0.1087	0.0165	0.6222	0.3034	0.3426
Speech input or realtime	0.0001	0	0	0.0008	0.0626	0.0014
Storage and egress	0.0001	0	0	0.0002	0.0001	0.0001
Total	0.306	0.110	0.0167	0.661	0.369	0.444
Narrated minutes	20.1	7.2	1.1	41.2	20.1 (+5)	22.7
Voice share of total	99%	99%	99%	94%	82%	77%

Table 1. Source: `benchmark/cost/cost_model.json` , price date 2027-01-01 for promotional prices. Realtime column uses gpt-realtime-2.1-mini; Gemini Live gives \$0.350 and the full gpt-realtime-2.1 gives \$0.496. The worst case is a policy upper bound (300 place queries an hour, maximum-length stories, six questions, three nearby searches), not a replay.

Per-session variable cost is almost all TTS

ESTIMATED list-price model, configured default mix (gpt-6-luna, gpt-4o-mini-tts, gpt-4o-mini-transcribe, Places Nearby Pro), 0% shared cache. Call counts MEASURED from deterministic replays. Prices retrieved 2026-09-27, evaluated at 2027-01-01. Model run 2026-10-04.

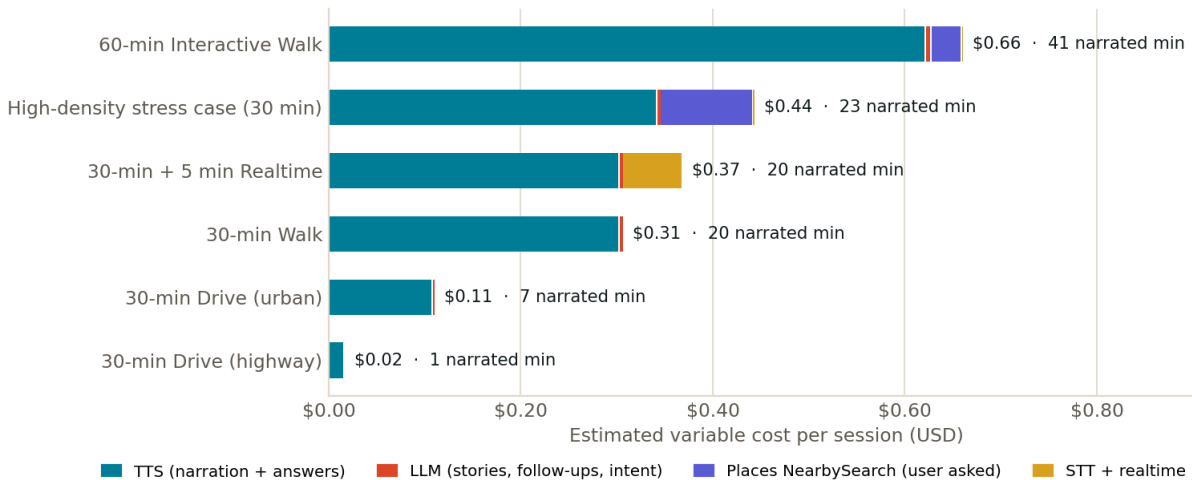


Figure 18. Per-session variable cost by component for each scenario and provider mix.

Data: `benchmark/cost/cost_model.json`. ESTIMATED list prices.

Where the money goes. Voice is 94% to 99% of variable cost in every walking or driving session without realtime and with at most one nearby search, and still 77% to 82% in the realtime and worst-case sessions. The language model is under 1% of a walk (0.9%); even a tenfold price increase on the model would add only about 9% to a walk. Two things follow. First, the product's economics are a *voice-rendering* problem, and the cure is to render each story once and share it (cache) or to render it on a cheaper voice (tier). Second, once voice is cheap, a different line appears: in the driver month with tiered voice and a 50% shared cache, Google nearby search is \$1.41 of \$2.96, about 48% of the bill.

12.3 Sensitivity to voice, model and maps pricing

Mix	30-min walk (0% / 50% cache)	30-min highway (0% / 50%)	What it shows
Configured default	0.306 / 0.161	0.0167 / 0.0090	Baseline
Tiered voice (highway on WaveNet)	0.306 / 0.161	0.0040 / 0.0021	Tier only affects the highway by default
All speech on WaveNet (sensitivity)	0.073 / 0.038	0.0040 / 0.0021	Walking at about 4x lower voice cost, if the voice is acceptable
OpenAI tts-1 (device speech input)	0.264 / 0.139	0.0143 / 0.0077	A cheaper OpenAI voice barely moves the result
Gemini stack at 2027 prices	0.374 / 0.197	0.0200 / 0.0110	About 22% above default after the promotion ends
Premium (Claude Sonnet 5 prose, ElevenLabs Flash voice)	0.924 / 0.486	0.0510 / 0.0270	About 3x default; breaks the \$1 per-session cap in a 60-minute walk
Risk: Google Places for automatic discovery	1.039 / 0.894	0.309 / 0.301	About 3.4x on a walk and 18x on the highway; Places content also cannot be pooled across users

Table 1. ESTIMATED, from `benchmark/cost/cost_model.json`. 50% shared-cache is an ASSUMED production rate.

A 25% change in voice price moves a walk by roughly 25% (about \$0.08), a 25% change in model price by under 0.3%, and moving automatic discovery onto Google Places multiplies cost several-fold. The design choice that protects the economics is therefore *free, shareable discovery* (Wikimedia) with Google paid only on a user question.

12.4 Long-haul driver month

Assumptions: 176 hours a month (8 hours, 22 days) on the I-40 replay rates, one question and a quarter of a nearby search per hour.
Revenue reference: a \$9.99 a month driver plan, which is a **HYPOTHESIS**, is \$8.49 net of a 15% store fee; the proposed cost target is under \$0.03 per active hour, which is \$5.28 a month.

Voice mix	0% shared cache	50% (ASSUMED)	80% (ASSUMED)	Per hour at 0%
Configured default	\$8.08	\$5.36	\$3.73	\$0.046
Tiered (highway on WaveNet)	\$3.61	\$2.96	\$2.57	\$0.021
All WaveNet	\$3.03	\$2.38	\$1.99	\$0.017
OpenAI tts-1	\$7.12	\$4.78	\$3.38	\$0.041
Gemini stack, 2027 prices	\$9.60	\$6.27	\$4.27	\$0.054
Premium	\$21.86	\$13.51	\$8.49	\$0.124
Risk: Places for discovery	\$110.84	\$108.12	\$106.49	\$0.630

Table 1. ESTIMATED. Source: `benchmark/cost/cost_model.json`, `truckDriverMonth`.

Long-haul driver month (176 h): tiered TTS + shared cache vs plan price

ESTIMATED list prices (2026-09-27, evaluated 2027-01-01); call rates MEASURED from the I-40 replay (2.85 stories/h, 18.25 free Wikimedia queries/h); after D-024 probe backoff; 1 question/h and 1 NearbySearch per 4 h ASSUMED. Plan price and store fee are HYPOTHESES. Model run 2026-10-04.

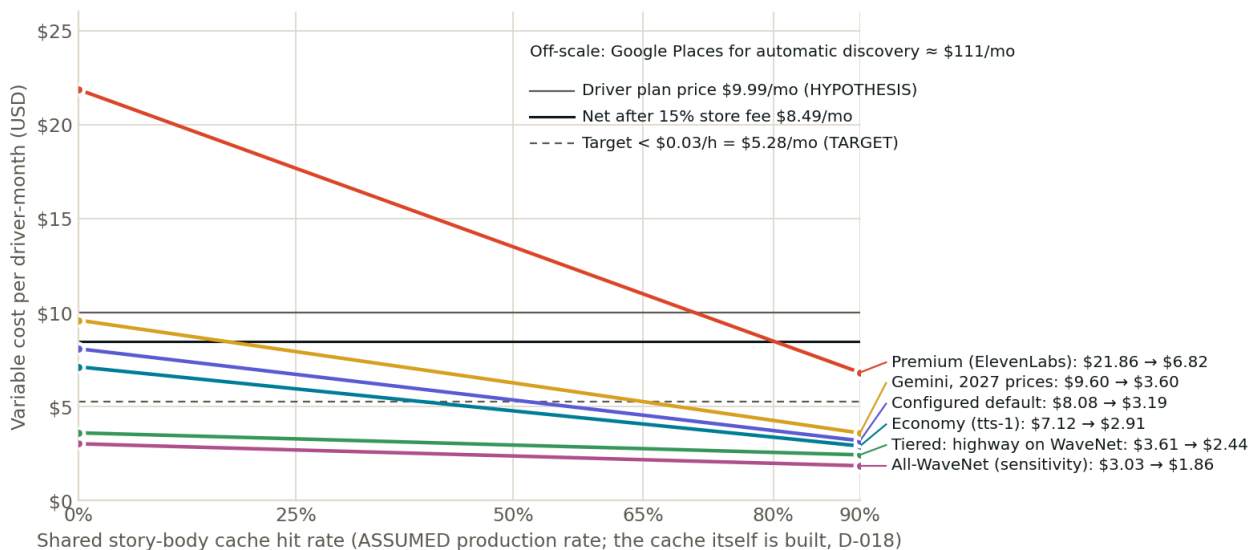


Figure 19. Driver-month cost against shared-cache hit rate for each voice mix, with the plan's net revenue and the per-hour target.

Data: `benchmark/cost/cost_model.json`. ESTIMATED; cache rates are ASSUMED.

On a pure sparse-highway month the plan clears the target only with the tiered voice, and it is fragile. On the default voice it is break-even at best (\$8.08 against \$8.49) and above the target even with a 50% cache (\$5.36 against \$5.28). With the tiered voice it clears the target at every cache rate on that month, but that month prices all 176 hours at the I-40 replay rate (2.85 stories an hour), a single synthetic and sparse trace. The sensitivity below recomputes the tiered-voice month at no cache from the same model. The real mix of road types and the real story rate are **NOT YET MEASURED**, and the result also depends on economy voices that listeners accept and a production cache hit rate (**NOT YET MEASURED**). The shared-cache key includes Guide, locale, angle, mode and length bucket, so reuse splits across them; a 50% hit rate would need heavily skewed place popularity and is untested.

Tiered voice, no cache	Month	Margin against \$8.49 net
0% urban hours (as modelled)	\$3.61	57%
10% urban hours	\$7.12	16%
25% urban hours	\$12.39	negative

Tiered voice, no cache	Month	Margin against \$8.49 net
Highway story rate 2.4 times higher (cadence bound)	\$6.64	22%

Table 1. ESTIMATED. An urban hour costs \$0.2201 on the default mix. Recomputed by the independent reviewer from the cost model’s own inputs (deliverables/REPORT_REVIEW.md , F-02).

12.5 Realtime active minute

Provider and model	Per active minute	Per 5-minute burst	Worst case, assistant talking all minute
Google gemini-3.8-live	\$0.0087	\$0.0435	\$0.018
OpenAI gpt-realtime-2.1-mini	\$0.0117	\$0.0624	\$0.024
OpenAI gpt-realtime-2.1	\$0.0369	\$0.1897	\$0.077

Table 1. ESTIMATED list price at 30% listener and 40% assistant audio per active minute, ten turns in five minutes, with OpenAI history re-billing counted as cached input. Realtime itself was NOT RUN. The agent-proposed budget is at most \$0.02 per active minute.

Worst case: if the assistant talks the whole minute, gpt-realtime-2.1-mini costs \$0.024 a minute, above the \$0.02 budget; the table shows typical values only.

12.6 Representative monthly cost

At the unit-economics assumptions in section 13 (25,000 monthly active users, 2.08 sessions per user per month, 34.75 minutes per session): about **\$14,600** a month of variable cost on the default mix with no shared cache, **\$7,950** at an ASSUMED 50% cache, and **\$7,900** with tiered voice at 50%, plus an ASSUMED \$450 of fixed infrastructure. Per user: a paying user at six sessions a month costs \$1.68; a free user at two sessions costs \$0.56. These are ESTIMATED costs at HYPOTHESIS usage.

13. Business model, go-to-market and economics

13.1 Options considered and the recommendation

Option	Fit with guest-first	Verdict
Freemium with a spoken-minutes cap	High: the first magic moment happens before any signup	Yes, as the base layer
Trip or day pass (non-renewing)	Very high: travellers dislike subscriptions	Yes, the primary offer for travellers
Annual plan	Needs a store identity	Yes, for repeat road-trippers and locals
Driver plan	Needs CarPlay or Android Auto and the safety benchmark	Phase 2, only if highway cost is proven
Ads or sponsored places	Hurts trust and conflicts with grounding and silence	No in version 1
White-label for tourism boards and attractions	n/a	Pilot as a channel, not a revenue pillar
Fleet driver-wellbeing	n/a	Pilot with one or two carriers after safety is proven
Vehicle-maker or rental licensing	n/a	Largest upside, longest cycle; Autio in Ford and Lincoln is the precedent

Tier	Initial price to test (HYPOTHESIS)	Includes	Logic
Guest, free	\$0, no account	Short trial sessions; both Guides	The proposal was about 30 narrated minutes and 5 questions a day, which the economics below show is unaffordable; see 13.3

Tier	Initial price to test (HYPOTHESIS)	Includes	Logic
Trip Pass	\$9.99 for 7 days	Unlimited narration and questions within fair use	Lower sticker price than Autio's 14-day \$14.99, but about 33% higher per day than Autio and above Waytale's 7- and 14-day passes (\$5.99 and \$9.99); matches how travellers buy
Explorer Annual	\$39.99 a year (monthly \$6.99 as a decoy)	Everything, year-round, journey history	Autio one-year range is \$35.99 to \$49.99; Roadtrippers Basic is \$35.99
Driver (phase 2)	\$9.99 a month or \$79.99 a year	Long-haul mode, silence controls	Needs highway cost under about \$0.03 an hour

Table 1. Prices are anchored on sourced competitor prices (retrieved 2026-09-27) and have **not** been tested with any customer. Source:

[docs/research/MONETIZATION.md](#).

Metering counts *spoken minutes only*, so silence is free and the product has no incentive to over-talk. Store fees are 15% on Google Play subscriptions and 15% to 30% on Apple (15% under the Small Business Program); web checkout savings are not planned for because Apple's link-out fee is in litigation.

13.2 Go-to-market and dependence

Channel	Why	Dependence and constraint
App Store and Google Play search	Intent keywords such as “road trip audio guide”	Crowded by scripted-tour apps rated 4.8 to 4.9 stars; a new app needs early ratings. Discovery is the dependence.
CarPlay	Road-trippers and truckers live in the car	Apple entitlement required; the voice-based conversation category requires voice as the primary mode with no text or imagery. Blocked today by the missing Apple Developer account
Android Auto	Same audience	No conversational category: the realistic path is a media-app integration with car-quality review
Travel creators (road-trip and van-life video)	The product demos well on video	Affiliate or revenue share; no on-screen interaction while driving
Road-trip planners and RV communities	Roadtrippers already bundles Autio	They can bundle us or treat us as a rival
Trucking communities and carriers	Isolation over long shifts (up to 11 driving hours)	Hand-held phone use is prohibited (fines up to \$2,750 for drivers and \$11,000 for employers); needs zero required touches and legal review
Tourism boards, rental fleets, vehicle makers	“Your destination, narrated everywhere”	Small today, long cycles; museums are a weak target because free alternatives exist

Acquisition cost is not modelled. No customer-acquisition cost, conversion funnel or store-conversion estimate exists, because no user has seen the product. The plan leans on organic store search, creators and partnerships rather than paid acquisition, and each of those is unproven. Launch sequence in the research (months 0 to 3 beta, 3 to 6 store launch, 6 to 12 driver beta, year 2 licensing) is a judgement, not a forecast; the dated roadmap in section 19 replaces it with gates.

13.3 Unit economics

Why the free tier is unaffordable as specified

Per monthly active user per month, USD; HYPOTHESIS revenue and usage, ESTIMATED cost (list prices)

Net revenue per MAU

\$0.059

HYPOTHESIS: 2.1% pay \$39.99/yr, 15% store fee

Variable cost, 50% shared-cache hit

\$0.317

ESTIMATED: 2.08 sessions x \$0.153; 50% is ASSUMED

Variable cost, no shared cache

\$0.582

ESTIMATED: 2.08 sessions x \$0.280

At the modelled usage a free user costs about 5.3 to 9.8 times the revenue they bring. Revenue funds only 0.21 sessions per MAU per month at the default voice mix; break-even needs 34% of users paying with no shared cache (the 2.1% freemium median is the hypothesis). Free access must be capped to a few trial sessions.

Source: benchmark/cost/cost_model.json unitEconomics (Medium case, 25,000 MAU); label: engineering model, not telemetry. Free-tier caps and Google free quotas ignored.

Figure 20. Why the free tier is unaffordable as specified: net revenue per monthly active user against variable cost per user, with and without a shared cache.

Data: benchmark/cost/cost_model.json unitEconomics, Medium case. Revenue and usage are HYPOTHESIS; cost is ESTIMATED.

- **A paying Explorer user is profitable on variable cost.** Net revenue is \$2.83 a month (\$39.99 a year less 15% store fee). Six sessions at \$0.280 each is \$1.68, a **41% variable margin** with no shared cache; at an **ASSUMED** 50% cache the cost is about \$0.92 and the margin about 68% (**DERIVED**). A 7-day Trip Pass nets \$8.49; at one 30-minute walk a day (an **ASSUMED** usage) the cost is about \$2.15, a margin of about 75% (**DERIVED**).
- **The free tier is not affordable as specified.** With 2.1% of users paying (the median freemium conversion in the research is a hypothesis for this product) revenue is \$0.059 per monthly active user, while the model's average user costs \$0.58 a month with no cache or \$0.32 at a 50% cache: **5.3 to 9.8 times the revenue they bring**. Revenue funds only 0.21 sessions per active user per month. Break-even needs roughly 34% to 36% of users paying with no shared cache. A free allowance of 30 narrated minutes a day, as first proposed, would cost about \$14 a month per user at the cap (**DERIVED**: 30 minutes, 30 days, about \$0.0153 per narrated minute), more than two hundred times that revenue.
- **The Trip Pass margin depends on light use.** The 75% margin assumes about 30 minutes of walking a day. At \$0.613 an hour with no cache, the \$8.49 net is used up after about 13.9 hours of walking, which is 2.0 hours a day over seven days (3.8 hours a day at an **ASSUMED** 50% cache); a three-hour day for seven days costs about \$12.87. The annual plan breaks even at about 55 hours a year with no cache (about 105 at 50%). "Fair use" must be defined before this price is tested.
- **What would fix it, in combination:** a free allowance of a few trial sessions rather than a daily allowance; a decision to put walking narration on the economy voice (about 4x cheaper voice, quality unbenchmarked); a real production cache hit rate (mechanism built, rate unmeasured); and trip passes, which this model ignores.

Scale case	Monthly active users	Variable cost: default, no cache	Variable cost: 50% cache	Fixed infrastructure (ASSUMED)	Revenue at 2.1% paying (HYPOTHESIS)
Small	1,000	\$583	\$318	\$60	\$59
Medium	25,000	\$14,585	\$7,952	\$450	\$1,487
Large	250,000	\$145,849	\$79,525	\$4,000	\$14,871

Table 1. At these usage and conversion hypotheses the blended business loses money at every scale; the unit economics only work for payers. Source: benchmark/cost/cost_model.json unitEconomics. Free-tier caps and Google free quotas are ignored.

Gross-margin implications. For payers the variable margin is healthy; for the whole user base the margin is negative until free usage is capped and the cache works. This is why the next experiments (section 19) measure cache hit rate and conversion before any scale spend, and why the free tier is the first thing to redesign.

Capital and operations required. Infrastructure is cheap for a pilot (an **ASSUMED** \$60 a month for one server, backups and a domain) and the spend that matters is variable cost per active user. Non-infrastructure requirements, none of which is priced here: trademark counsel, an Apple Developer account and a CarPlay entitlement application, Google Play review, legal review of driver-safety claims and of Google Places terms, a listening test with raters in English and Russian, and field-test time on real roads. No dollar figure is claimed for these.

14. Product metrics and admin model

The first-party admin console is built on the API's metric endpoints, which are SQL over the product's own events, sessions, latency samples and cost ledger. Role-gated access (analyst, admin, owner) and an audit log cover every admin action.

View	What it shows	Why it matters	Template area
Overview	Daily, weekly and monthly active users (guest and account), sessions, mean duration, stories funnel (offered, started, completed, skipped, interrupted, resumed, abandoned), questions, nearby searches, Guide mix, regime mix	Is anyone using it, and how far do they get	Activation, engagement
Latency	p50 and p95 per interaction: trigger to first audio, speech end to first audio, barge-in stop, nearby search	The product lives or dies on responsiveness	Performance
Cost	Spend by provider, task and day; cost per session and per active user, from the price table	The unit-economics risk in section 12 is visible as it happens	Unit cost
Providers	Calls, errors, fallbacks, cache hit counters by layer, circuit-breaker states, configured providers	Detects outages and the cache hit rate that decides viability	Provider reliability, unit cost
Quality	"Not that one" taps, skips, ratings, grounding failures	Wrong-target and boredom signals	Story quality
Geo	Coarse geohash aggregates with a minimum of five sessions per cell	Where it is used, without tracking individuals	Engagement
Session explain	The decision timeline for one session: scores and suppression reasons	Answers "why did it say that, or not"	Story quality
Budgets and kill switches	Per-category limits and switches such as "disable the language model" (owner only, audited)	Cost containment without a deploy	Unit cost
Health and audit	Database and cache status, recent errors, build identifier; append-only audit log	Operations and accountability	Provider reliability

Table 1. Source: `docs/API.md` admin table and `apps/api/src/routes/admin.ts`.

Measured data versus illustrative data. The metric queries are exercised by an automated test (8 passing tests), and the console screens in **Figure 7** were captured with the *demo data* toggle on: every number in those images is synthetic. No production or field data exists, so the console has never displayed a real measurement. **Retention readiness** is partial: a stable guest identifier is kept for up to 90 days and sessions are stored, so cohort retention can be computed, but no cohort or retention view is built. There is no signup funnel either, because the product is guest-first; activation is measured as the first story started. The mobile admin surface (health, sessions, latency, cost, errors, device list) is implemented but has never run on a device.

15. Acceptance benchmark

The owner's benchmark defines 27 scenarios in sections A to F, with the realtime voice scripts as D-B1 to D-B6. Each is marked below with evidence from the run of 2026-10-04. Status meanings: PASS means the demonstrable part passed in tests, replay or the harness; PARTIAL means that part passed but a live, on-device or human-rated check is missing; NOT RUN means it needs credentials, devices or raters. **Nothing is hidden, and nothing failed outright.**

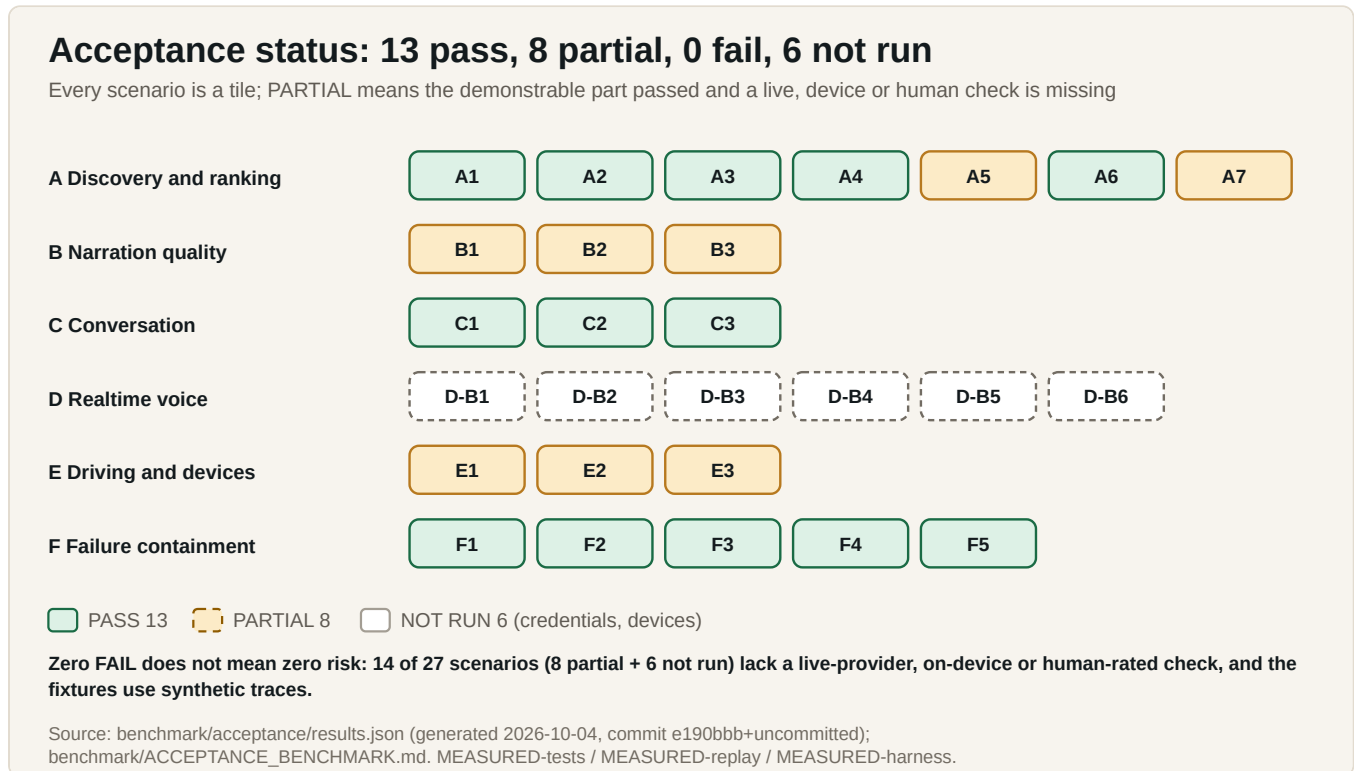


Figure 21. Acceptance status: 13 pass, 8 partial, 0 fail, 6 not run.

Data: benchmark/acceptance/results.json (generated 2026-10-04, commit label e190bbb+uncommitted; the work has since been committed as 51a787e).

15.1 All 27 scenarios

ID	Scenario	Status	Evidence and what is missing
A1	World Trade Center / Ground Zero	PASS	9/11 Memorial is the first story at 361 m; all local WTC targets precede distant ones; reversed provider order gives identical selection (MEASURED-replay)
A2	Golden Gate Bridge	PASS	Bridge is the first story, triggered at 5,764 m (reach 12,160 m) against small-place reach; Vista Point suppressed with reasons
A3	Art Institute of Chicago	PASS	Museum first at 426 m; Willis Tower, Field Museum and Navy Pier never precede it; the enclosing city is ambient context
A4	Weak evidence	PASS	Thin evidence gets at most one fact-free orientation line, never a story; 0 grounding failures; invented names and years are rejected
A5	Highway ahead-discovery (truck)	PARTIAL	I-40 replay: 5 stories, 0 behind, 0 passed mid-story, 98.7% silent in replay (about 96% with model-length stories, section 10.3), longest gap 32 min, 18.2 place queries an hour; truck stops, off-route and behind candidates rejected with reasons. No real-road run
A6	Environment transition	PASS	Highway to urban to stationary to walking in one session, 0 restarts, no retold targets; walking look-ahead under one tenth of highway; questions only on foot
A7	Cross-city portability	PARTIAL	Same runner and fixture source in New York, Chicago and San Francisco with no per-city configuration; source scan finds no city coupling. Live Wikimedia discovery not run (sandbox has no access)
B1	Short then deeper	PARTIAL	Follow-up draws only unspoken facts from the same pack; with nothing left it says so. Model non-repetition not run

ID	Scenario	Status	Evidence and what is missing
B2	Guide differentiation	PARTIAL	Same lead fact for both Guides in 31 of 31 packs; persona parameters differ. Prose difference and human rubric not run
B3	Callback	PARTIAL	Callbacks come only from structured journey memory with shared tags; model coherence not run
C1	Coffee interruption	PASS	36 of 36 harness sessions: stop before answer, story preserved, validated map, grounded answer, resume at the same plan; about \$0.035 per coffee turn (ESTIMATED)
C2	Contextual follow-up	PASS	Subject resolved without restatement; no place, evidence or nearby call (36 of 36); answer grounded against the active brief (fake model)
C3	Change topic	PASS	Explicit topic change abandons the story; the old plan never replays over 20 more frames
D-B1	Realtime: story interruption and coffee	NOT RUN	Credentials not supplied
D-B2	Realtime: contextual follow-up	NOT RUN	Credentials not supplied
D-B3	Realtime: barge-in “No, I meant parking”	NOT RUN	Not run on a realtime provider; the hybrid-path analogue passes
D-B4	Realtime: road noise	NOT RUN	Credentials and a noise fixture needed
D-B5	Realtime: Guides and languages rubric	NOT RUN	Human raters and credentials needed
D-B6	Realtime: session lifecycle	NOT RUN	Server side exists; no provider session can be opened
E1	Driving and OTR safety	PARTIAL	Drive HUD with one mic target and no text input; highway stories at most 75 s; speech held during manoeuvres. No vehicle or device run
E2	Background and sleep resilience	PARTIAL	Event-driven flush and background buffering unit-tested; OS limits documented. No device run
E3	Network degradation	PARTIAL	Exactly-once resume, no tool re-run, no story restart (tests). On-device playback during an outage not run
F1	Empty discovery result	PASS	4 place-source calls in 300 frames; empty 105-minute highway replay at most 40 calls
F2	Place provider quota or timeout	PASS	Timeouts: 3 attempts per 300 frames; 429 and 401: exactly 1 attempt
F3	Model failure	PASS	Failing or hallucinating model gives a template story for the same target; at most 2 model calls per story; kill switch gives 0
F4	Voice failure	PASS	Segments and answers delivered as text; at most 4 voice attempts; conversation continues
F5	Cache unavailable	PASS	Readiness reports degraded and reduced mode; 1 place call per 300 frames; narration continues

Table 1. Source: `benchmark/acceptance/results.json` . Counts: 13 PASS, 8 PARTIAL, 0 FAIL, 6 NOT RUN.

Zero FAIL does not mean zero risk. Fourteen of 27 scenarios (8 partial plus 6 not run) lack a live-provider, on-device or human-rated check, and every PASS runs on synthetic traces and fake providers. A skeptical reader should read the acceptance result as “the logic behaves as designed”, not “the product works in the world”.

15.2 Summary metrics (the owner’s section G)

Metric	Result	Label
Target relevance regression pass rate	26 of 26 replay relevance assertions (A1 to A3, A5 to A7)	MEASURED-replay
Wrong-target rate on the fixed corpus	0 of 33 stories started (behind, passed mid-story, clutter or utility targets) across 6 runs of 5 distinct synthetic traces (the I-40 trace runs twice, so 28 distinct starts); assertions and fixtures were written by the build agents and not held out	MEASURED-replay , 6 synthetic scenarios
Weak-evidence correctness	0 grounding failures; 3 orientation-only lines; 0 stories on thin evidence	MEASURED-replay
Narrative repetition regressions	0 retold targets in any replay; model paraphrase repetition not measured	MEASURED-replay / NOT YET MEASURED
Guide and persona distinctness	Brief-level facts identical; prose distinctness not measured	NOT YET MEASURED

Metric	Result	Label
Interruption correctness	36 of 36 harness sessions stop, answer, resume the same plan	MEASURED-harness
Contextual follow-up grounding	Grounded in the test; 36 of 36 made no place search (fake model)	MEASURED-tests
Trigger to first audio p50 / p95	1,663 / 2,683 ms (n = 36, loopback)	MEASURED-harness , simulated providers
Barge-in stop p50 / p95	1.0 / 1.5 ms server round trip; device not measured	MEASURED-harness
Speech end to first audio p50 / p95	Nearby 1,250 / 2,328 ms; follow-up 1,950 / 3,380 ms	MEASURED-harness , simulated providers
Provider tool reliability	Not measured on real providers; invalid fake rows dropped 36 of 36	NOT YET MEASURED
30-min walk, drive and realtime minute cost	\$0.306; \$0.110 urban, \$0.0167 highway; \$0.0117 per minute	ESTIMATED
Cache hit rates by layer	Production not measured; harness values in section 8	MEASURED-harness / NOT YET MEASURED
Known failures	None in automated evidence; open risks in section 18	MEASURED-tests

15.3 Cost and performance budgets, including the misses

The agent-proposed budgets (the template has no numbers) are scored separately from acceptance, and some are missed. They are shown because a reader should see them.

Budget	Target	Actual (label)	Status
30-min walking cost	at most \$0.15	\$0.306 default; \$0.161 at an ASSUMED 50% cache; about \$0.07 on an economy voice (ESTIMATED)	FAIL by default
30-min urban drive cost	at most \$0.10	\$0.110 (ESTIMATED)	FAIL, slightly above
Highway cost per active hour	under \$0.03	\$0.033 default; \$0.008 tiered (ESTIMATED)	Pass only with the tiered voice
Realtime cost per active minute	at most \$0.02	\$0.0117 and \$0.0087 (ESTIMATED ; realtime not run)	PARTIAL
Follow-up latency	2.0 s / 3.5 s	1.95 s / 3.38 s (was 2.84 / 4.92 s, a fail)	PARTIAL: simulated, 3% margin
Native startup	1.5 s / 3.0 s	none	NOT RUN
Failure-loop containment	no unbounded loop	0 unbounded; every failure test bounded to 1 to 4 calls	PASS

Table 1. Source: docs/PERFORMANCE_COST.md section 17.

16. Code quality, security and operations

16.1 Tests and build

Suite	Tests	Kind	Status
Core decision package	189	Unit and property tests; includes a source scan that forbids city names and fixture imports	189 pass
Providers	104	Adapters (recorded-shape fixtures, streaming), router, guard, pricing	104 pass
Replay and acceptance	35	Deterministic replays over synthetic traces, relevance assertions	35 pass
Shared client	29	Sequencer, reconnect, batching, prefetch, local engine	29 pass
API	88	Integration against real Postgres and Redis with fake providers: conversation, admin, resilience, cost and latency	88 pass
Mobile logic	15	Pure helper logic only; no UI or device tests	15 pass

Suite	Tests	Kind	Status
Total	460		0 failures
Web end-to-end (Playwright, Chromium, offline demo)	5	Home, no horizontal scroll at 390 px, offline walk shows the 9/11 story, and others	5 pass

Table 1. Source: `benchmark/acceptance/results.json` suites (2026-10-04). Every provider is a fake in these tests. The code base is about 28,000 lines of TypeScript source and about 5,000 lines of tests (counted 2026-10-04; blank lines and comments included). Type checking is strict and clean.

CI. A GitHub Actions workflow is defined: type check, all test suites against Postgres and Redis service containers, the replay suite, the provider benchmark in no-key mode, and a container image build on `main`. **No CI run has ever executed:** the repository has not been connected to a runner. The workflow as written does not run the web end-to-end suite. All results here are local runs.

16.2 Security and privacy

Area	What exists	Gap
Secrets	None in the repository; <code>.env</code> and <code>API-Credentials.md</code> are git-ignored (the latter does not exist here); the logger redacts tokens, keys, cookies and precise coordinates, and strips tokens from URLs; the reverse proxy drops authorization headers from access logs	Production secrets management is by operator process only
Authentication	Guest-first JWT (90 days, no personal data); optional email code login only when an email provider is configured; admin by WebAuthn passkeys; phone admin by revocable device pairing	No external security review; no penetration test
Authorisation	Roles owner, admin, analyst; every admin mutation is appended to an audit log	n/a
Rate limiting	600 requests a minute per address globally; stricter limits on guest creation (20), login (10), nearby search (30), speech upload (30) and realtime token (10); per-session and per-minute provider caps; a \$1.00 estimated spend cap per session	Process-local budget windows (see scalability)
Data minimisation	Precise position kept in memory and a short Redis window; Postgres stores a coarse geohash (about 5 km) and regime only; no raw traces unless the user opts in; speech clips deleted immediately	Opt-in “save my journey” path not exercised by users
Retention	Events 13 months; cost ledger and audit log 25 months; place history 90 days; deletion of a guest or account on request	Retention job is scheduled by the operator (not yet deployed)
Dependency vulnerabilities	pnpm audit on 2026-10-04 reports 16 advisories, none remediated: 2 critical, 5 high, 9 moderate. Runtime: the web map library (critical, an HTML sanitiser bypass). Build and test tooling only: the test runner (critical), PostCSS (three, high), the bundler, a certificate library and a glob library (high)	Exploitability in this application has not been assessed; upgrades have not been attempted
Driver safety	No required touches while driving; no Guide questions; listening only on an explicit press; speech held during manoeuvres	Not validated in a vehicle; legal review of hand-held-use rules not done

16.3 Operations

- **Deployment.** One server running Docker Compose (reverse proxy with automatic certificates, API, web, Postgres, Redis). The compose file and Dockerfile exist; **they have never been built on a real host**, and SSH deploy from CI is a TODO. The API itself boots and passes its tests locally.
- **Observability.** Health and readiness endpoints (readiness reports degraded, reduced mode and circuit-breaker states), a first-party metrics pipeline and the admin console. No external error monitoring or alerting is configured.
- **Backup and rollback.** Documented: a nightly database dump with 14-day rotation (audio is a rebuildable cache), and rollback by redeploying the previous image tag, with forward-only additive migrations so the old image still works. Neither has been executed.
- **Public, mobile and admin deployment status.** None is deployed. See section 17.

17. Implementation status and demo readiness

State	What
Implemented and tested (with fakes)	Decision core (regimes, corridors, director, safety, resume, brief, grounding, segments, intent, memory); provider router and guard with budgets and breakers; API with sessions, directive stream with exactly-once resume, conversation tools, nearby search, auth and roles, admin metrics, retention; story-body cache, tiered voice routing, streamed follow-ups; replay harness; WebApp, public website and admin console (build and run locally); mobile app code (Explore, Drive HUD, History, Settings, onboarding, simulation, operator screens)
Partially implemented or simulated	Offline demo mode runs the real core over fixture places (labelled); admin console shows demo data only; mobile bundles compile for both platforms and expo prebuild succeeds, but no native build exists; Google Cloud voice adapter is implemented but its voices are unaudited candidates; Wikimedia adapter is implemented but was never run live
Blocked by missing credentials or access	Realtime voice benchmark (OpenAI and Google keys); real model, voice and speech-to-text runs (keys); live nearby search (Google Places key) and Google maps in the apps (Maps keys); Android build (Expo token, Android SDK); iOS build (Apple Developer account, a macOS host); CarPlay and Android Auto (entitlements); public deployment (server and domain with DNS); email login (email provider); live discovery testing (Wikimedia is unreachable from this sandbox)
Not implemented	CarPlay and Android Auto integrations; wake word; cohort retention view; offline route pre-download; creator or partner tooling; any B2B console; payments and subscriptions

URLs and builds. *Public URLs for the WebApp, website and admin console: not deployed. Blocked on server and domain credentials.* No installable native build exists. No URL is given here because none exists.

To see it today. The website, WebApp and admin console can be run locally from the repository in offline demo mode (`pnpm install` , then build and start `apps/web` ; no keys are needed), and the 74-second recording is in the packet. **To install on a phone** once credentials arrive (from `docs/MOBILE.md`): supply an Expo token and run an internal-distribution build with the `preview` profile to get an Android APK that can be installed directly; for iPhone, an Apple Developer membership is needed for an ad hoc build (devices registered first) or TestFlight. Set the production API address in the build, otherwise the app runs only in labelled offline demo mode.

Not evidenced anywhere in this report: any real user or listener; any run on a device; any live provider call; real map rendering; real voice quality; the mobile admin surface; a real-road field run; production cache hit rate; public deployment; a legal trademark position.

18. Risks and critical assumptions

PART IV · RISKS, PLAN AND ASSESSMENT

Category	Risk	Impact	What we know	Mitigation or next step
Market	Big-platform risk: Google, Apple or OpenAI add proactive narration to an installed, free assistant	Existential	Platforms answer questions free today; no platform feature we found narrates unprompted stories about what you pass, although Google describes Gemini in navigation as offering recommendations “based on your route”, so the line between answering and proactive is already thin (last searched 2026-09-27); Google judged medium-high within 12 months (our estimate)	Verticals (long-haul, partners, fleets), editorial quality, journey memory; monitor releases
Market	Willingness to pay for narration is unproven; Q&A is free	High	Scripted tours sell at \$15 to \$40; AI-native lookalikes have negligible traction	Charge for proactive, curated narration and memory, not Q&A; test pass price
Market	Crowded category; a direct AI audio-city-guide competitor (Waytale, live since 2025-09, cheaper passes) and several others are already live	High	Waytale listing read 2026-10-04; others by title	Differentiate on silence policy, trajectory reasoning and grounding, and prove it with users
Product	Walking cadence may be too talkative (about two thirds narrated with model-length stories)	Medium	Derived from the cost model, section 10.3	Quieter default; measure with listeners
Product	No user has judged timing, relevance or voice	High	Zero real users	Closed beta
Technical	Nothing has run on a device, a road or a real network	High	Bundles compile only	Device and field runs

Category	Risk	Impact	What we know	Mitigation or next step
Technical	Live discovery untested: Wikimedia is unreachable here; real candidate density and rate limits unknown	High	Rate limit of 5,000 requests an hour per token caps roughly 32 to 90 concurrent sessions per address	Live runs in three cities; self-hosted Wikidata and OSM extract
Technical	Synthetic traces and hand-set significance values drive every replay result	High	Fixtures are hand-curated, not sampled from real places	Real-road runs; sampled fixtures
Provider	No real model, voice or realtime provider exercised; model names conflict between sources	High	Section 9	Provider benchmark; model-list check
Provider	Google Places terms: cache limits and Google-map-only display; the display rule is not enforced in code	High	Section 9	Enforce or change source before live Places traffic; legal review
Provider	Gemini promotional prices double on 2027-01-01	Low	Modelled at the 2027 price	Provider swap by configuration
Cost	Voice is 94% to 99% of variable cost; free tier unaffordable as specified	High	Section 12 and 13	Tiered voice, shared cache, capped free tier
Cost	Production cache hit rate unknown; the 50% and 80% rates are assumed	High	Harness 100% is a single repeated route	Measure in beta
Cost	Economy voices unaudited; walking-tier decision open	Medium	Quality unknown	Listening test, English and Russian
Safety, privacy, security	Driver distraction and hand-held-use rules; legal exposure for carriers	High	Design avoids required touches; unvalidated in a vehicle	Closed-course test; legal review
Safety, privacy, security	9 unremediated dependency advisories (1 critical)	Medium	Section 16	Upgrade and re-audit
Safety, privacy, security	Brand not cleared; domains unregistered; Telvey is a working name	Medium	Section 6	Register domains; clearance search
Scalability	Sticky WebSockets, local audio, process-local budget windows block multi-instance	Medium at scale	Documented triggers (about 20 concurrent sessions)	Redis-backed windows, object storage and CDN

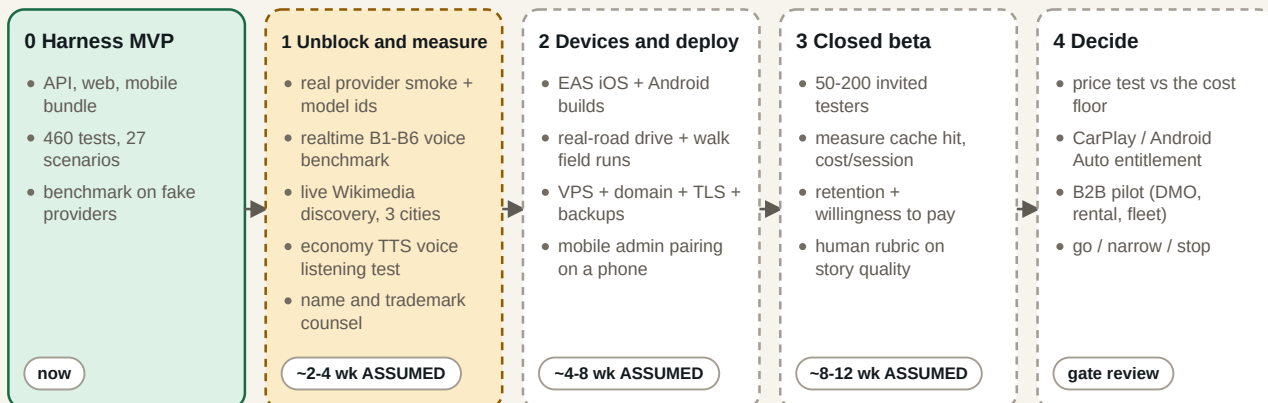
Assumptions requiring real-world validation: that people want unprompted narration; that the shared story body will be re-used often enough (at least 50% of plays); that listeners accept an economy voice on highways; that long silences feel right rather than broken; that the guest-first free allowance can be capped without killing conversion; that drivers will pay about \$9.99 a month; that the 120 million US road-tripper figure and the 2% penetration are in the right region; and that platforms leave the proactive niche open long enough to matter.

19. Roadmap, milestones and funding logic

The plan is a sequence of gates. Each gate removes a named blocker or tests one assumption, and none promises a date or a budget. Durations are planning assumptions that depend mostly on how fast credentials arrive.

Roadmap: a sequence of gates, not a calendar

Each phase removes a named blocker or tests one assumption; durations are ASSUMED



Funding logic: what each scenario is meant to prove (no precise budget is claimed)

Bridge (phases 1-2)

Proves the product works with real providers, on real devices and on a real server. Output: a measured cost per session and a latency table replacing the estimates.

Beta (phase 3)

Proves people use it twice and that the shared cache and tiered voice bring cost under the plan price. Output: first real hit rate, retention, conversion signal.

Scale (after phase 4)

Only if the beta clears the cost floor and a channel (stores, OEM, fleet, DMO) is identified. Not proposed on current evidence.

Source: this report's synthesis of docs/STATUS.md blockers (D-010, D-016) and benchmark gaps. Durations are planning assumptions, not commitments; no dates are promised.

Figure 22. Roadmap as gates and funding logic. Durations are ASSUMED planning estimates, not commitments.

Source: this report's synthesis of docs/STATUS.md blockers (D-010, D-016) and the benchmark gaps.

Gate	Removes or tests	Evidence that would pass it (PROPOSED, not validated)
1. Unblock and measure (about 2 to 4 weeks, ASSUMED)	Provider smoke test and model identifiers; realtime benchmark; live Wikimedia discovery in three cities; economy-voice listening test; trademark counsel and domain registration	Realtime table filled; grounding pass rate at least 95% after one retry; 95th-percentile latency inside the targets on real providers; economy voice rated at least 3.5 of 5
2. Devices and deployment (about 4 to 8 weeks, ASSUMED)	Android and iOS builds; real-road drive and walk field runs; server, domain, certificates and backups; phone admin pairing; Google-map display rule enforced	Both builds installed on real phones; an I-40 style field run with recorded selected targets and silence; a deployed URL with a restore test
3. Closed beta (about 8 to 12 weeks, ASSUMED)	50 to 200 invited testers; first real cache hit rate, cost per session, retention and willingness to pay; human rubric on story quality	Shared-body hit rate near or above the assumed 50%; measured cost per session consistent with section 12; a return-rate threshold agreed before the beta (no category benchmark exists)
4. Decide	Price test against the cost floor; CarPlay and Android Auto entitlements; one partner pilot (tourism board, rental, carrier)	A go, narrow or stop decision on the evidence

Funding logic: what each scenario is meant to prove. No budget is claimed, because the drivers are unknowns that gate 1 and 3 will measure.

- **Bridge (gates 1 and 2).** Proves the product works with real providers, on real devices and on a real server. The output is a measured cost per session and a measured latency table that replace today's estimates. It is mostly engineering time, credentials and small provider spend.
- **Beta (gate 3).** Proves that people use it twice and that the shared cache and tiered voice bring cost under what a user pays. The output is the first real hit rate, retention and a conversion signal. The main cost is variable provider spend for the testers plus time to

run listening tests.

- **Scale (after gate 4).** Only if the beta clears the cost floor and a distribution channel (stores, vehicle makers, fleets or tourism partners) is identified. **Not proposed on current evidence.**

20. Final assessment

Why it could matter. Place knowledge is abundant and free; what is scarce is *judgement about when to speak*, and the willingness to stay silent. A companion that tells a short, true story about the thing you are passing, answers your question from verified data, and otherwise leaves you alone is a different product from a chatbot you have to query. Long-haul drivers, who spend hours in sparse country with nobody to talk to, and travellers who pay \$15 to \$40 for scripted tours, are real populations with real spending on adjacent products.

What the implementation demonstrates. The central design is coherent and testable: a deterministic brain that decides what to say, when and whether it is safe; a model confined to writing sentences that are checked against evidence; exact interruption and resume; provider independence; bounded failure behaviour with no unbounded paid loop; and a cost architecture (shared story body, tiered voice, free shareable discovery) aimed at the real cost driver. It does so with 460 passing tests, a 27-scenario benchmark with no failures and a clear map of where the evidence stops. Cost and latency problems were found by measurement and fixed (follow-up latency, highway cost, density-probe churn), which is a good sign about the process.

What still must be proven before meaningful scale or investment. Everything that touches the world. That anyone wants it; that it works on a phone, on a road, with real providers and real data; that real stories are grounded and sound good in the Guides' voices; that the cache hit rate and economy voices bring cost under what users pay, which the model says is the whole question for drivers; that platforms do not make the niche disappear; and that the name can be used. The free tier must be redesigned before launch, because as specified it costs 5 to 10 times the revenue it earns.

The next decisive experiment. Supply the blocked credentials and, in one pass, (1) run the provider and realtime benchmark, (2) put the build on two phones and drive a real highway and walk a real downtown, and (3) hand the app to 50 to 200 people for a few weeks while the admin console records cache hit rate, cost per session, return rate and “not that one” taps. If the cache hits at 50% or better, the economy voice passes a listening test and people come back, the thesis survives its first contact with reality. If not, the cheapest failure is the one that happens first.

21. Comparison packet

A single folder and archive, `telvey-comparison-packet.zip`, accompanies this report. It holds the report in PDF and as editable source (this Markdown and the builder script); the 16 template screenshots (15 captured; the mobile admin surface is listed as not captured) with an index; the brand board, logos and app icons; Guide portraits, scenes and avatars; the 13 architecture, market and economics figures and five charts as SVG and PNG; benchmark and cost-model data as JSON and CSV; the mobile and deployment guides (`docs/MOBILE.md` , `docs/DEPLOY.md`); the 74-second demo recording; a README that indexes everything; and a manifest with a SHA-256 hash for every file.

Public URLs for the WebApp, website and admin console: not deployed. Blocked on server and domain credentials. Native test-build distribution: no build exists; the instructions are in `docs/MOBILE.md` , section 5, and summarised in section 17.

Appendix A. Evidence labels and sources

Label	Meaning
MEASURED-tests	An automated test ran in this build (vitest or Playwright), with deterministic fakes wherever a provider is involved
MEASURED-replay	A deterministic replay of a synthetic trace over fixture places and evidence through the production decision code
MEASURED-harness	The real API (Fastify, Postgres, Redis, WebSocket) in-process on loopback, with fake providers and simulated latency
MEASURED-e2e	Playwright against the production web build in offline demo mode
ESTIMATED	A list-price model, with prices retrieved 2026-09-27 (promotional prices evaluated at 2027-01-01)
ASSUMED , HYPOTHESIS	An input or belief with no measurement behind it; always stated with its rationale
DERIVED	Arithmetic on the numbers above, shown in the text
NOT YET MEASURED , NOT RUN	Needs credentials, devices, field runs or human raters that were unavailable

Principal external sources (all retrieved 2026-09-27 unless noted; headlines are context only and are not used in the bottom-up model): U.S. Travel Association forecast (domestic trips and leisure spend); AAA Memorial Day 2026 forecast (39.1 million car travellers); U.S. Bureau of Labor Statistics, heavy and tractor-trailer drivers (2.22 million jobs, May 2025); FMCSA hours-of-service and mobile-phone rules; Menlo Ventures 2026 consumer-AI survey (64% of adults use AI, 55% of those pay; the \$40 billion headline); RevenueCat 2026 subscription benchmarks (2.1% median freemium conversion at day 35); Edison Research Infinite Dial 2026 (in-car listening); Destination Marketing Association International (658 member organisations; a Wikipedia infobox figure whose body text says 575); Verified Market Reports (the \$2.86 billion audio-tour headline; methodology not visible); competitor app-store listings and websites for Autio, GuideAlong, VoiceMap, Shaka Guide, SmartGuide, izi.TRAVEL, Narrativ, RoadWhisper and others; Forbes on Autio in Ford and Lincoln vehicles (2026-08-04); Apple and Google store-fee pages; Google Maps Platform pricing and service-specific terms; OpenAI developer pricing and model pages; Google Gemini API and Cloud text-to-speech pricing; Anthropic pricing; Deepgram, ElevenLabs, Cartesia and Mapbox pricing; Wikimedia API rate-limit page (marked draft). Full URLs and access dates are in `benchmark/research/*.json` and `docs/research/*.md`. Search-result titles on 2026-10-04 (unverified): Waytale, Tourly Audio, CityTour AI, Guideline, Narriva and Votura. Platform announcements were not re-researched on 2026-10-04. RDAP and web screening for the brand: 2026-09-27, 16:15 to 16:35 UTC.

Appendix B. Corrections to earlier documents and known inconsistencies

These are statements in earlier project documents or earlier drafts of this report that did not survive a closer check, and what was done.

#	Earlier statement or state	Correction
1	<code>docs/STATUS.md</code> (2026-09-27) says the core is in progress and replay, backend, web and mobile are not started	Stale; everything exists. <code>docs/STATUS.md</code> was rewritten to the current state (this is the one earlier file edited)
2	An early draft of the competitor matrix scored Telvey's <i>target</i> features as if built	Now two rows: "as built" (what works in tests, replay or harness) and "target" (aspiration); section 3
3	The competitor research (19 entries) did not include Waytale; the brand screen had rejected the <i>name</i> Waytale for collision	Waytale added to the landscape as a live direct competitor found 2026-10-04 (not otherwise verified)
4	OpenAI's pricing page on openai.com and its developer docs listed different flagship model names	Developer docs used; identifiers to be confirmed by model-list call; section 9
5	<code>docs/PERFORMANCE_COST.md</code> : voice is "95% to 99% of every walking or driving session"	True for most; 94% in the 60-minute interactive walk and 77% to 82% in the realtime and worst-case sessions; in the tiered driver month nearby search is about 48% of cost; section 12
6	Replay reports of 84% to 91% silence on walks and 98.7% on the highway read as listener experience	They are measured on template-length speech; with model-length stories walking is about 33% silent and the highway about 96%; section 10.3

#	Earlier statement or state	Correction
7	<code>benchmark/acceptance/results.json</code> is labelled commit <code>e190bbb+uncommitted</code>	The work was committed afterwards as <code>51a787e</code> ; the file is generated and was not edited
8	Docs state that Places content must be shown on a Google map, implying it is handled	It is documented only: the web map defaults to OpenStreetMap and iOS uses Apple Maps without a Google key; no code enforces it; section 9
9	No earlier document mentions dependency advisories	<code>pnpm audit</code> reports 16 (2 critical); section 16
10	Domain availability was reported “free”	True at 2026-09-27 16:15 to 16:35 UTC; not re-verified on 2026-10-04 (lookup tool denied); may have changed
11	The cost model’s “price date” is 2027-01-01	That is the date for promotional-price evaluation, not a retrieval date; retrieval is 2026-09-27
12	MONETIZATION.md proposed about 30 free narrated minutes a day as cost-bounded	The cap bounds cost at about \$14 a month per free user, which is more than 200 times modelled revenue per user; the free tier must be redesigned; section 13.3
13	Earlier report text overstated the source line count, and a later recount understated it	Recounted 2026-10-04 from tracked files: about 28,000 lines of TypeScript source and 5,000 of tests; section 16

Appendix C. Questions an investor should ask

1. Has anyone used it? No. Zero real users, zero device runs, zero live-provider runs. Every performance figure is from real code against fake providers with simulated latency.

2. What stops Google, Apple or OpenAI from doing this next quarter? Nothing structural. Q&A about places is already free and installed; proactive narration is one setting away. We think Google is the most likely to close the gap and rate it medium-high within 12 months, which is our own judgement. The defensible positions are narrow: long-haul drivers, partner channels and execution quality.

3. What is actually different, today? A deterministic policy for *when* to speak (including long silence), trajectory-ahead reasoning, grounded text, exact resume and a cost architecture. All are built and tested on synthetic data; none is validated by users, and competitors can copy each.

4. Are the stories true? Every spoken sentence is checked against approved evidence and unsupported numbers, years and names are rejected. That has been tested with *fake* prose. No real model’s output has been checked. The evidence source (Wikipedia and Wikidata) has not been reached from our environment, so live evidence quality is untested.

5. What does it cost per user, and who pays? About \$0.31 per 30-minute walk and \$0.017 per 30-minute highway drive on default settings, 94% to 99% of it voice rendering (**ESTIMATED**). A payer at six sessions a month leaves about a 41% variable margin. A free user costs \$0.56 a month at two sessions against about \$0.06 of revenue per user. The free tier as originally specified is unaffordable and must be redesigned.

6. Does the truck driver plan work? Only with the tiered voice plan and a shared cache. At default settings a driver month costs \$8.08 against \$8.49 of net revenue; with the tiered voice it is \$3.61, and \$2.96 at an assumed 50% cache. Neither the highway-tier voices (still pending benchmark) nor the cache hit rate has been measured.

7. How big is the market? On a bottom-up model, about \$5.6 billion TAM, \$1.9 billion SAM and about \$6.5 million of year-three revenue (range \$3.7 to \$13 million) across three consumer segments, all **ASSUMED** . The year-three figure is “what a fraction of one percent would pay”, not a forecast, and the road-tripper population (120 million) has no sourced count.

8. What would make you stop? A cache hit rate far below 50%, an economy voice that testers reject, a return rate near zero in the beta, or a platform shipping proactive narration. The roadmap is built so the cheapest of these fail first.

9. Is the name yours? Not yet. Telvey is a working name; the screening was preliminary and non-legal; the domains are unregistered; a direct competitor, Waytale, appeared for a name we rejected.

10. Is it safe to use while driving? The design avoids required touches and Guide questions while driving, but it has not been tested in a vehicle or reviewed legally. Hand-held phone use is prohibited for truck drivers, so a carrier would require both.

11. Do the maps and places data terms work? Partly. Google Places data must be shown on a Google map and cached only briefly. Our code avoids caching and uses a free source for discovery, but the Google-map display rule is not enforced and the web and iOS defaults would violate it if live Places data were shown.

12. How secure is it? Passkey-only admin login, role checks, audit log, no secrets in the repository, rate limits and redaction. But there has been no security review, and `pnpm audit` shows 16 unremediated advisories including two critical (one in the runtime web map library, one in the test tooling).

13. Who built this, and who has reviewed it? AI coding agents built the code, tests, fixtures, benchmarks, cost model and this report for one product owner. No independent human has reviewed the code, tests or security; the same agents wrote the tests and the code, so passing tests prove internal consistency only. This is a key-person and verification risk, not a team. A human engineering and security review is part of gate 1.

13. Can it scale? Not yet and not claimed. The first ceiling is Wikimedia's rate limit (tens of concurrent sessions per address), then single-process WebSockets and local audio. Fixes are known and not built.

14. Are the passes honest? 13 pass, 8 partial, 0 fail, 6 not run. Fourteen of 27 lack a live, device or human check. Two team-set cost budgets (30-minute walk and urban drive) are currently missed.

15. What is the first thing you would fund? Credentials and a few weeks to run the realtime benchmark, real devices and a real road, followed by a small closed beta. The report does not ask for money for scale and does not claim a budget.

Appendix D. Reproducing the numbers

What	Command or file
All tests	<code>pnpm -r test</code> (needs Postgres 16 and Redis 7 for the API suite)
Acceptance benchmark (about 7 minutes)	<code>pnpm bench:acceptance</code> writes <code>benchmark/acceptance/results.json</code> and <code>latency.json</code>
Cost model	<code>pnpm bench:cost</code> writes <code>benchmark/cost/cost_model.json</code>
Replays	<code>pnpm replay</code> ; outputs in <code>benchmark/replay/</code>
Provider benchmark (needs keys)	<code>pnpm bench:providers</code>
Report numbers check	<code>python3 scripts/report/check-numbers.py</code> asserts that the headline numbers in this report appear, as formatted strings, somewhere in the report and match the benchmark JSON; it does not check where they appear, derived figures (for example margins and break-even hours), audit counts, external facts or figure text
Screenshots and demo	<code>scripts/report/capture-screenshots.mjs</code> , <code>scripts/report/record-demo.mjs</code>
Figures	<code>scripts/report/figures_arch.py</code> , <code>scripts/report/figures_data.py</code>
This report	<code>python3 scripts/report/build_report.py</code> (Markdown to HTML to PDF with Chromium)